

Defending Digital Democracy: Mitigating Redundancy Attacks in Voting Advice Applications

Submission 239

Abstract. Voting Advice Applications (VAAs) match voters with candidates based on policy preferences, but their default recommendation algorithms weight all survey questions equally. Malicious actors can exploit this vulnerability by injecting redundant or correlated questions to artificially amplify specific policy dimensions and skew recommendations. We propose an automated defense that assigns non-uniform weights to survey questions in the standard distance-based recommendation algorithm. These weights are derived from the clone-robust weighting (CRW) axiomatic framework, which operates on an embedding of questions in a metric space and reduces the influence of redundant or strategically injected questions. We instantiate the framework with off-the-shelf sentence embeddings and evaluate it on SmartVote data from the 2023 Swiss National Council elections, selecting the embedding model and radius on the canton of Zurich and testing on Bern, with a distinct electorate and candidate pool. With E5-Instruct, CRW improves robustness to synthetic clones in 98.9% of question–attack pairs, mitigating 17–47% of the induced distortion, and offsets 40–63% (low-effort) and 20–30% (high-effort) of artificial party-visibility gains under targeted attacks, while leaving recommendations on the unmanipulated questionnaire nearly unchanged. These results support CRW as an automated first-line defense, and open the door to instantiations based on response-pattern similarity—an arguably broader notion of redundancy.

Keywords: Voting Advice Application · Clone-robust Weighting · Semantic Redundancy · Sentence Embedding Models · SmartVote Dataset.

1 Introduction

Voting Advice Applications (VAAs) match voters with candidates or parties based on their answers to a policy questionnaire. Deployed in over 30 countries and consulted by up to 50% of voters [25], they measurably affect both turnout and vote choice [18]. In Switzerland, SmartVote was consulted by up to 20% of eligible voters in the 2023 National Council elections [21], and 67% of its users report that it influenced their vote [13,12]. When a single recommender shapes the information environment of a fifth of the electorate, the integrity of its algorithm is a matter of direct democratic consequence.

Most VAAs, including SmartVote, invite voters to weight questions by importance. In practice, however, this input is often coarse, and a majority of users do

not provide any weights at all. In such cases, the platform defaults to computing voter–candidate distances with metrics that weight all questions equally [3]. This creates a structural vulnerability: if several questions measure the same policy dimension, whether by accident or by design, that dimension gains disproportionate influence over the recommendations. A question designer can exploit this by adding paraphrased or topically redundant questions, amplifying a favored issue without any single question appearing suspicious. Although this vulnerability has been identified [10,3], its impact and mitigation remain unexplored.

We propose to replace uniform weights with *clone-robust weights* (CRW) [4,5]. Given a metric over questions, this axiomatically grounded framework assigns weights that vary continuously with the questions’ positions and are inversely proportional to their local density, so that adding near-duplicates only redistributes weight locally and a tight cluster of redundant questions collectively receives the weight of a single isolated one. Its effectiveness, however, depends entirely on the space used to measure similarity, a choice especially critical in a democratic context: should distances reflect population responses, capturing culturally specific structure but potentially obscuring minority viewpoints; or should they minimize such population biases and rely solely on semantic similarity? Negation adds a further complication: “Should taxes be raised?” and “Should taxes be lowered?” are semantically opposite, yet encode the same policy dimension, and a defense that fails to treat them as redundant leaves a substantial attack surface open.

Contribution. This work provides the first systematic empirical validation of defenses against redundancy attacks in questionnaire-driven recommendation systems. Our contribution is threefold. First, we formalize question redundancy in VAAs and assess the structural vulnerability it induces on SmartVote data, showing that even a minimal redundancy attack measurably distorts voter–candidate recommendations. Second, we adapt CRW to the questionnaire setting and clarify its design space, showing how the choice of representation, distance, and interaction radius determines the notion of redundancy corrected. Third, we evaluate a semantic instantiation of this defense against synthetic clones of varying difficulty. We find that CRW consistently improves robustness across embedding models, hyperparameters, attack settings and cantons, mitigating 17–47% of induced distortions, and reduces artificial party-visibility gains by up to 63% under low-effort attacks.

2 Related Work

Voting Advice Applications. VAA recommendations depend heavily on design choices: the distance metric [16] and the selection of statements [27] alone can change which candidates are recommended, while how these choices shape recommendations is rarely disclosed to users [10]. Berdoz et al. [3] catalogue 11 manipulation strategies on SmartVote data and single out *question correlation* as particularly insidious: unlike candidate-level attacks, it can be exploited by a

single question designer and affects all voter–candidate pairs at once. They leave its quantification and mitigation open; we address both.

Handling Redundancy. Redundant items distorting aggregate outcomes is a recurring problem: social choice addresses it through independence of clones [26] and false-name-proofness [8], model evaluation across overlapping benchmarks faces the same issue [2,17], and repeated training data measurably harms language models [7]. The simplest remedies make hard, binary decisions about redundancy: *deduplication* removes items whose embeddings lie within a similarity threshold of one another [1], while *cluster normalization* partitions items into clusters and weights each cluster as a single item [9]. Both are discontinuous: a small perturbation of the items can flip whether an item counts as redundant, or move it to another cluster, abruptly changing its weight and that of its neighbors. More recent work seeks smoother alternatives. Procaccia et al. [22] weight alternatives in preference learning according to their Voronoi regions, an approach that nonetheless fails some natural robustness guarantees [4]. Clone-robust weighting (CRW) [4,5] addresses these limitations by deriving weights from symmetry, continuity, and locality axioms: weights vary continuously with the items’ positions, and adding a near-duplicate only redistributes weight among its neighbors, so that a tight cluster jointly receives the weight of a single isolated item. The graph-based construction proposed in [5] averages cluster-style weights over a range of thresholds instead of committing to one, making CRW a continuous and local generalization of threshold-based approaches. To our knowledge, CRW has not yet been applied to recommendation data.

Sentence Embeddings. We obtain question distances from transformer sentence encoders. *STS models* such as SBERT [23] are trained to place sentences of similar meaning close together, and thus tend to separate a statement from its negation. *Retrieval models* such as E5 [28], Jina-v3 [24], GTE [15], and Qwen3 [30] instead match on topic regardless of stance, which is desirable here since a question and its negation measure the same policy dimension. Fine-tuning SBERT on political data can further capture the correlation structure of beliefs [14], a natural route toward behaviorally grounded distances.

3 Problem Setup & Defense Mechanism

VAA Model. We model a VAA as a mapping from the policy positions of voters and candidates to a personalized ranking of candidates for each voter. Let $\mathcal{Q} = \{q_t\}_{t=1}^Q$ denote the set of questions, $\mathcal{V} = \{v_i\}_{i=1}^V$ the set of voters, and $\mathcal{C} = \{c_j\}_{j=1}^C$ the set of candidates. Each question q_t admits a finite set of answers $A_t \subseteq [0, 100]$, and each individual $x_i \in \mathcal{V} \cup \mathcal{C}$ is represented by its response profile $\mathbf{x}_i = [x_{i,1}, \dots, x_{i,Q}]^\top \in \prod_t A_t$, where $x_{i,t} \in A_t$ denotes the answer of x_i to q_t . Additionally, each voter v_i may assign personalized importance weights $w_i \in \Delta(\mathcal{Q})$ to the questions; when none are provided, the system defaults to uniform weights $w_t^{\text{unif}} := 1/Q$. For each voter v_i , candidates are then ranked

by increasing weighted ℓ_p distance $d_p(\mathbf{v}_i, \mathbf{c}_j, \mathbf{w}_i) = (\sum_{t=1}^Q w_{i,t} |v_{i,t} - c_{j,t}|^p)^{1/p}$, $p \geq 1$, yielding a ranking r_i which we view interchangeably as a total order over $\mathcal{C}$ and as a map $r_i : \mathcal{C} \rightarrow \{1, \dots, C\}$. Note that SmartVote is a special instance of this formulation, with $p = 2$ and unnormalized weights $\tilde{w}_{i,t} \in \{0, 0.5, 1, 2\}$, set to 1 by default, applied as $(\sum_t (\tilde{w}_{i,t} (v_{i,t} - c_{j,t}))^2)^{1/2}$, which is equivalent to setting $w_{i,t} \propto \tilde{w}_{i,t}^2$ since normalization does not affect rankings. Questions can equivalently be viewed as the pair of response vectors they induce over voters and candidates, i.e., $q_t := (q_t^V, q_t^C) \in \mathbb{R}^{V+C}$. Assuming uniform weights w^{unif} for all voters, the ranking $r_i(\mathcal{Q})$ is then fully determined by the questionnaire $\mathcal{Q}$.

Approximate Question Clones. To capture redundancy, one may consider questions in vector form $q_t = (q_t^V, q_t^C)$ and equip $\mathbb{R}^{V+C}$ with a notion of distance. Note however that this space exhibits additional structure induced by negation: if respondents were fully consistent, the answers to “Should taxes be raised?” could be inferred from those to “Should taxes be lowered?”. We hence define the involution $\sigma : x \mapsto 100 - x$ extended component-wise, and consider pseudo-metrics¹ d^{beh} that are invariant under σ , i.e., $d^{\text{beh}}(x, y) = d^{\text{beh}}(x, \sigma(y))$, such as $d_p^{\text{beh}}(x, y) = \min\{d_p(x, y), d_p(x, \sigma(y))\}$, where d_p denotes the standard ℓ_p distance. For a threshold $\alpha \geq 0$, questions q and q' are then α -behavioral clones if $d^{\text{beh}}(q, q') \leq \alpha$.

However, this view admits limitations. First, even logically equivalent questions, such as exact negations, may elicit different answers in practice, e.g., due to acquiescence bias [11]. Second, behaviorally close questions may still concern distinct policy topics, their proximity being driven by shared response patterns among a majority of respondents (e.g., carbon taxation and public transport investment). We therefore generalize the notion of clones by mapping questions into a representation space $\mathcal{E}$ via an embedding function $E : \mathcal{Q} \rightarrow \mathcal{E}$. Equipped with a pseudo-metric $d^{\mathcal{E}} : \mathcal{E} \times \mathcal{E} \mapsto \mathbb{R}_{\geq 0}$, the space $\mathcal{E}$ can be interpreted either as an idealized policy space or as a practical instantiation, e.g., via sentence embeddings. For a fixed embedding function E and threshold $\alpha \geq 0$, we then similarly define α -embedding clones as questions q and q' such that $d^{\mathcal{E}}(E(q), E(q')) \leq \alpha$.

Redundancy Attacks. An adversary transforms the original questionnaire $\mathcal{Q}$ into $\mathcal{Q}' = \mathcal{Q} \cup \mathcal{Q}_{\text{clone}}$, where $\mathcal{Q}_{\text{clone}}$ contains ℓ α -behavioral clones of each of n target questions in $\mathcal{Q}$, so that the VAA computes rankings $r_i(\mathcal{Q}')$ instead of $r_i(\mathcal{Q})$. To see why this matters, consider a single target q with ℓ exact (0-behavioral) clones: for voters without personalized weights, the system defaults to w^{unif} , and the effective weight of q increases from $\frac{1}{Q}$ to $\frac{1+\ell}{Q+\ell}$, i.e., by a factor of roughly $1 + \ell$ when $Q \gg \ell$. The same effect persists for α -behavioral clones with small α : a well-intentioned designer may thus inadvertently distort recommendations by including questions with similar response patterns, while a strategic one may deliberately inject approximate clones of questions favorable to a targeted party.

¹ A pseudo-metric relaxes the *identity of indiscernibles* of a metric to $d(x, x) = 0$, allowing distinct points at distance zero.

At the individual level, we measure the severity of an attack by the *Jaccard similarity* between the top- k candidate sets $T_k(\cdot)$ before and after manipulation, $J_k(r_i(\mathcal{Q}), r_i(\mathcal{Q}')) := \frac{|T_k(r_i(\mathcal{Q})) \cap T_k(r_i(\mathcal{Q}'))|}{|T_k(r_i(\mathcal{Q})) \cup T_k(r_i(\mathcal{Q}'))|}$, and complement it with *Spearman's rank correlation* ρ_S over the full candidate list. At the population level though, what matters is the electoral impact on each party. Following [3], we define the k -*visibility* of party p as the average fraction of top- k slots occupied by its candidates $\mathcal{C}_p \subseteq \mathcal{C}$, i.e., $\nu_k(p, \mathcal{Q}) := \frac{1}{V_k} \sum_{v_i \in V} |T_k(r_i(\mathcal{Q})) \cap \mathcal{C}_p|$, and its *visibility distortion* as $\Delta\nu_k(p) := \nu_k(p, \mathcal{Q}') - \nu_k(p, \mathcal{Q})$.

CRW Defense Mechanism. We replace uniform weights with clone-robust weights, adopting the graph-based construction of [5]. Let $S = \{E(q_t)\}_{t=1}^Q$ be the set of embedded questions, and $G_r(S)$ the r -neighborhood graph with vertex set S and edges between all pairs of questions within distance r according to $d^\mathcal{E}$. Given a probability density ν supported on $[0, \alpha]$ and a *clone-robust graph weighting function* w —ensuring weight-sharing across neighbors only [5, Def. 2]—each question receives the weight $f_{\nu, w}(S)(q_t) := \int_0^\alpha \nu(r) w(G_r(S))(q_t) dr$. Though abstract, this construction is intuitive: graph weighting w acts like a cluster normalization at each radius r ; maximal radius α thus controls the locality of weight sharing, as questions farther than α from an added clone remain essentially unaffected; finally, rather than committing to a single threshold whose crossing would abruptly change weights, aggregating over radii makes weights Lipschitz-continuous [5, Thm. 1].

We use the simplest instantiation, with a uniform density $\nu_\alpha(r) = 1/\alpha$ on $[0, \alpha]$ and the class-uniform graph weighting function w^{CU} of [5]. The latter groups nodes with identical closed neighborhoods, i.e., $u \equiv_G v$ if $N_G(u) \cup \{u\} = N_G(v) \cup \{v\}$, assigns equal mass to each equivalence class, and distributes it uniformly within classes: $w^{\text{CU}}(G)(u) := (|V/\equiv_G| \cdot |[u]_G|)^{-1}$, where $[u]_G$ is the class of u and $V/\equiv_G$ the set of classes. This yields prior weights $w_t^{\text{CRW}}(\mathcal{Q}) := f_{\nu_\alpha, w^{\text{CU}}}(E(\mathcal{Q}))(q_t)$, computable in $O(Q^4)$ time [5, Thm. 1], which remains negligible for questionnaires of up to several hundred questions. We denote by $r_i^{\text{CRW}}(\mathcal{Q})$ the induced rankings, and by ν_k^{CRW} and $\Delta\nu_k^{\text{CRW}}$ the corresponding party-level measures; the impact of an attack is then assessed by comparing $r_i^{\text{CRW}}(\mathcal{Q})$ with $r_i^{\text{CRW}}(\mathcal{Q}')$.

While the defense mechanism accommodates diverse design choices, any concrete instantiation requires specifying (i) the embedding function E , (ii) the distance $d^\mathcal{E}$, and (iii) the radius α . The choice of E and $d^\mathcal{E}$ is clearly critical, as it determines which questions are deemed redundant and thus which redundancies CRW corrects. The radius α must in turn balance locality: too small, and CRW reverts to uniform weights in the absence of exact clones; too large, and all questions collapse into a single equivalence class, again recovering uniform weights. In the remainder, we instantiate and evaluate a semantic-based defense, where E embeds questions based solely on their text, and treat α as a hyperparameter.

4 Experimental Setup

Datasets. SmartVote is Switzerland’s primary VAA, used in the 2023 National Council elections by up to 20% of eligible voters and 85% of candidates [21]. We use its 2023 data, collected and anonymized by Politools in accordance with Swiss data protection law and preprocessed following [3], for two cantons: Zurich ($V = 41,658$ voters, $C = 1,029$ candidates) and Bern ($V = 34,052$, $C = 685$). Both share the national questionnaire of $Q = 75$ questions, with answer sets $A_t = \{0, 25, 75, 100\}$ for 60 policy questions, $\{0, 17, 33, 50, 67, 83, 100\}$ for 7 value questions, and $\{0, 25, 50, 75, 100\}$ for 8 budget questions. All hyperparameters are selected on Zurich and then frozen; Bern, with a distinct electorate and candidate pool, serves exclusively as test set. Since SmartVote prominently displays as many candidates as there are seats in the canton, we evaluate the Jaccard similarity J_k and party visibility ν_k with $k = 36$ for Zurich and $k = 24$ for Bern. Spearman’s ρ_S leads to the same conclusions (rank correlation of 0.965 with J_k across questions and attack sizes) and is reported in the extended version.

Embedding Models. We instantiate E with ten off-the-shelf sentence encoders spanning semantic-similarity, retrieval, and instruction-tuned models: SBERT [23], E5 and E5-Asym [28], E5-Instruct and E5-Asym-Instruct [29], BGE-M3 [6], GTE [15], Nomic-v2 [19], Jina-v3 [24], and Qwen3-Embedding [30]. Instruction-tuned models receive a short directive framing the task as identifying the political topic of a question regardless of stance. Distances are measured with the normalized Euclidean distance $d^E(q, q') = \|\hat{E}(q) - \hat{E}(q')\|_2 = \sqrt{2(1 - \cos \theta_{q,q'})}$, where $\hat{E}(q) = E(q)/\|E(q)\|_2$ and $\theta_{q,q'}$ is the angle between embeddings, consistent with the cosine-based objectives these models are trained on.

Synthetic Clones. For each question, we prompt GPT-4o [20] to generate four instances of four clone types of increasing difficulty: *easy paraphrases*, with substantial lexical overlap; *hard paraphrases*, sharing no content words with the original; and *easy* and *hard negations*, obtained by negating the corresponding paraphrases. For instance, “Should busy sections of highways be widened?” yields the easy paraphrase “Should congested areas of highways be expanded?”, the hard paraphrase “Is there a need to increase the breadth of heavily trafficked motorway segments?”, the easy negation “Should busy sections of highways remain the same width?”, and the hard negation “Would it be advisable to maintain the current width of major roads in areas with substantial vehicle presence?”. All clones were manually checked to address the same policy issue as their original. Clones are assumed to be 0-behavioral, i.e., they inherit the answers to the original question, transformed by σ for negations. From these clones, we define five attack settings injecting four clones per target question: one *full* setting per clone type, and a *mixed* setting with one clone of each type.

Reproducibility. Clones are generated with `gpt-4o-2024-08-06` at temperature 0.8. All experiments run on an Intel Xeon E5-2690 (4 cores, 32GB RAM); each evaluation (two recommendation runs over all voters) takes about two minutes,

while CRW computations are negligible. Since voter-level data are private but available from Politools upon request [21], we release all code, generated clones, prompts, embeddings, and CRW weights in the extended version.

5 Results

We organize our results around six research questions: *RQ1*: How severely do redundancy attacks distort recommendations? *RQ2*: Which embedding model and radius α provide the best defense? *RQ3*: Does the selected configuration generalize to unseen questions, voters, and cantons? *RQ4*: Does CRW mitigate attacks targeting a specific party? *RQ5*: Does CRW remain effective when clones are only approximately behavioral? *RQ6*: What is the impact of adopting CRW on a clean questionnaire?

RQ1: Severity of Redundancy Attacks. For each question, we inject $\ell \in [1, 30]$ exact copies (0-embedding and 0-behavioral clones) and measure the resulting distortion under uniform weights on Zurich. A single copy already measurably alters recommendations for every question ($J_{36} \leq 0.977$), and distortion grows monotonically with ℓ for all questions. At $\ell = 4$ clones, J_{36} ranges from 0.657 for the most vulnerable question (retirement age) to 0.928 for the most robust one (security spending), so that redundancy is a structural vulnerability of the questionnaire rather than an artifact of a few polarizing topics (per-question results in the extended version).

RQ2: Model and Radius Selection. To maximize the signal for model comparison, we target five of the most affected questions in RQ1 on Zurich and, for each of the five attack settings, inject four clones per target question ($|\mathcal{Q}_{\text{clone}}| = 20$). We then sweep α over 21 values in $[0.01, 3]$, aggregate each robustness measure across voters and attack settings, and select for each model the radius α^* maximizing a composite score over these measures. Without defense, these attacks reduce J_{36} to 0.452, displacing nearly half of each voter’s top-36 candidates. As shown in Figure 1, CRW improves robustness for all ten models and radii, though their response to α varies structurally: the best models, E5-Instruct (peak $J_{36} = 0.584$) and Jina-v3 (0.581), exhibit sharply peaked profiles, whereas weaker models such as SBERT or E5-Asym-Instruct (0.490) remain flat over wide ranges of α . Spearman’s ρ_S yields the same ranking, improving from 0.905 to 0.952 for E5-Instruct. We hence select E5-Instruct with $\alpha^* = 0.4$, which maximizes its composite score ($J_{36} = 0.576$), and freeze this configuration for all subsequent experiments.

RQ3: Generalization. We next attack every question of the Bern questionnaire individually, injecting four clones under each attack setting, and benchmark CRW against the uniform baseline using the configuration selected in RQ2. Sweeping α on the full questionnaire first confirms that the frozen radius remains near-optimal: all clone types peak at $\alpha = 0.3$ or 0.4 , and $\alpha^* = 0.4$ yields

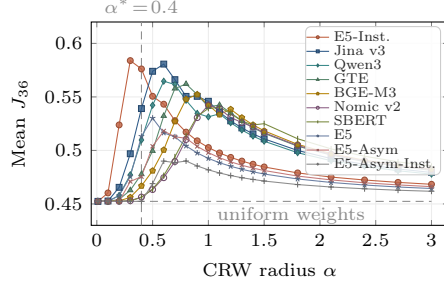


Fig.1: Mean Jaccard similarity as a function of CRW radius α for all 10 embedding models, averaged over all five attack settings. The dashed horizontal line marks the uniform-weight baseline ($J_{36} = 0.452$). Models are listed in the legend by composite rank.

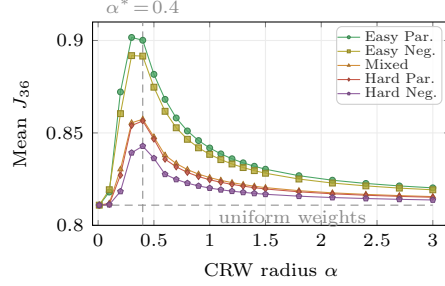


Fig.2: Mean Jaccard similarity as a function of CRW radius α across all 75 questions, for each clone type (E5-Instruct). The dashed horizontal line marks the uniform-weight baseline ($J_{36} = 0.811$). All clone types peak at $\alpha = 0.3$ or 0.4 , close to the selected α^* .

the best aggregate performance (Figure 2). At this radius, CRW improves robustness in 371 of 375 question–attack pairs (98.9%), the remaining four showing negligible degradation (< 0.002 ; Figure 3): the defense thus transfers from the five selection questions in RQ2 to the full questionnaire, despite the change of population and candidate pool. The mitigated share of distortion follows attack difficulty, from 47% for easy paraphrases and 42% for easy negations, to 24% for mixed settings and hard paraphrases, and 17% for hard negations. This hierarchy mirrors clone generation: paraphrases are easier to detect than negations, and hard variants lie farther from their originals in embedding space.

RQ4: Targeted Party Manipulation. We focus on the six major parties—the Social Democratic Party (SP), the Green Party (Green), the Green Liberal Party (GLP), The Centre (Centre), FDP. The Liberals (FDP), and the Swiss People’s Party (SVP)—which collectively span the Swiss political spectrum and account for the vast majority of SmartVote usage. We consider an omniscient white-box adversary who knows both the response data and the recommendation algorithm, and thus provides an upper bound on achievable manipulation. For each party, the adversary identifies the five questions whose duplication most increases that party’s visibility ν_{24} , and injects either one easy paraphrase of each (*low effort*, 5 clones) or one clone of each type (*high effort*, 20 clones). As shown in Table 1, both attacks yield systematic gains, of up to +2.16pp and +5.07pp respectively. CRW consistently reduces these gains, by 40–63% under low-effort attacks (mean 55%) and by 20–30% under high-effort attacks (mean 28%), the latter including the hardest clone types. Importantly, the correction operates jointly across parties: the targeted party’s reduced gain is accompanied by a recovery of the parties it took visibility from. Under the high-effort attack favoring the Centre,

for instance, CRW reduces its gain from +3.44pp to +2.46pp, while SP’s loss shrinks from −3.09pp to −2.46pp and the Green’s from −2.20pp to −1.64pp.

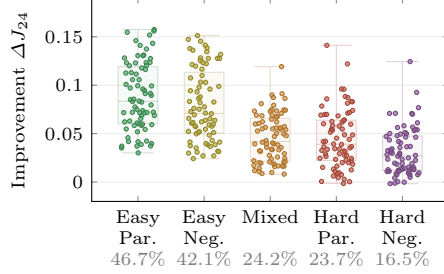


Fig. 3: Per-question CRW improvement ΔJ_{24} on Bern across 75 questions and 5 attack settings (E5-Instruct, $\alpha^* = 0.4$). Gray percentages indicate the mean share of distortion mitigated $\overline{\Delta J_{24}}/(1 - \overline{J_{24}})$, with uniform baseline $J_{24} = 0.810$ across all conditions.

Party	Low Effort ($\ell = 1$)			High Effort ($\ell = 4$)		
	$\Delta\nu_{24}$	$\Delta\nu_{24}^{\text{CRW}}$	Mitig.	$\Delta\nu_{24}$	$\Delta\nu_{24}^{\text{CRW}}$	Mitig.
SP	+2.16	+1.05	51.6%	+5.07	+4.06	19.8%
GLP	+1.64	+0.66	59.8%	+4.18	+2.94	29.7%
Green	+1.32	+0.57	57.1%	+3.26	+2.27	30.3%
Centre	+1.30	+0.78	40.3%	+3.44	+2.46	28.6%
FDP	+0.38	+0.15	60.8%	+1.03	+0.73	29.4%
SVP	+0.26	+0.10	63.0%	+0.82	+0.58	29.9%
Mean	+1.18	+0.55	55.4%	+2.97	+2.17	27.9%

Table 1: Visibility gain $\Delta\nu_{24}$ (pp) of each targeted party on Bern relative to the unmanipulated questionnaire, under uniform weights and CRW (E5-Instruct, $\alpha^* = 0.4$), with $n = 5$ target questions per party. “Mitig.” gives the share of the gain mitigated by CRW.

RQ5: Approximate Behavioral Clones. In practice, clones need not induce exactly the same answers as their original. Since measuring how respondents actually answer clones would require a costly new survey, we simulate behavioral drift with synthetic perturbations. We inject clones under the mixed setting and perturb their answers: each voter’s and candidate’s answer to each clone independently switches with probability λ to another value of A_t , drawn with probability proportional to $|a_{\text{new}} - a_{\text{old}}|^{-1}$. For $\lambda > 0.2$, clones already correlate less with their original than the most correlated pair of distinct SmartVote questions ($r = 0.78$), covering a broad range of approximate clones. As shown in Figure 4, robustness degrades gracefully as clones drift from their behavioral target, with J_{36}^{CRW} decreasing from 0.82 at $\lambda = 0$ to 0.73 at $\lambda = 1$. Even at $\lambda = 1$, where every answer is changed, robustness does not collapse: since nearby answers are favored, clones remain weakly correlated with their original ($r = 0.13$). Throughout, CRW keeps a stable advantage of about 0.07 over uniform weights, showing that the defense does not rely on clones inducing identical answers.

RQ6: Impact on Clean Questionnaires. Finally, we apply CRW to the unmanipulated questionnaire to assess its influence in the absence of attacks. Weights deviate from uniform by only −16% to +3%, the largest reductions affecting topically overlapping pairs such as electricity-market liberalization and progressive electricity tariffs (−16% each), or health-insurance deductibles and premium subsidies (−14% and −11%). Accordingly, recommendations on Bern remain

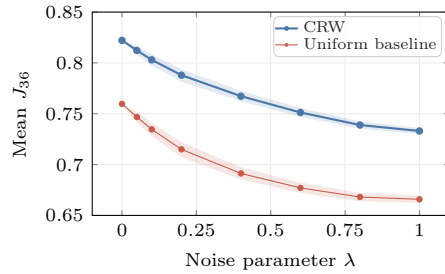


Fig. 4: Robustness to approximate behavioral clones under the mixed setting, where each voter’s and candidate’s answer to each clone switches with probability λ . Each point averages J_{36} over all voters, 75 questions and 20 repetitions; shaded bands indicate the 90% confidence interval.

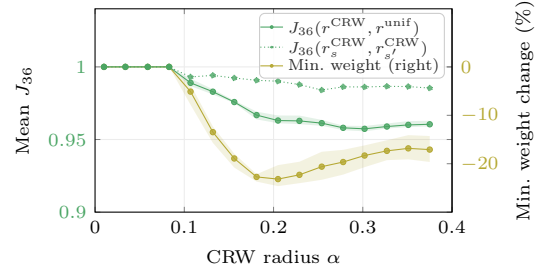


Fig. 5: CRW with behavioral distance d_1^{beh} , computed from all candidates and a random 5% voter subsample (seed s); J_{36} is averaged over the remaining voters, and yellow gives the largest weight reduction vs. uniform. Bands give the range over 5 seeds; α ranges up to the 75th percentile of question distances.

largely unchanged: only 0.87 of the top-24 candidates are swapped out on average ($J_{24} = 0.93$), and party visibility shifts by at most 0.13pp. Adopting CRW would thus amount to a minor correction, which reflects the careful design of the SmartVote questionnaire: its questions are well separated in embedding space and present little semantic redundancy, leaving CRW little to correct in the absence of adversarial manipulation.

6 Discussion

Main Findings. Our results confirm that the vulnerability identified in [3] is potent: a single clone already measurably distorts recommendations, with effects growing monotonically with additional clones. The semantic defense evaluated in Section 5 consistently improves robustness across embedding models, radii, clone types, questions, cantons, and behavioral noise, mitigating 17–47% of the induced distortion, and substantially reduces artificial party-visibility gains under targeted attacks. On the unmanipulated questionnaire, CRW only marginally alters weights and recommendations, so that its adoption comes at little cost—validating the careful design of the SmartVote questionnaire.

Meaning of Redundancy in VAAs. Our defense only addresses a narrow, text-based notion of redundancy. First, semantic embeddings are inherently vulnerable to evasion: sufficiently complex clones—double negations, convoluted paraphrases, or jailbreak-style rewrites—can always bypass them. Second, text similarity may miss deeper logical dependencies: expanding the army may be tightly coupled to increasing the defense budget, and positions on environmental issues may largely reduce to prior beliefs about anthropogenic climate change. Both

limitations suggest response patterns as a more robust basis for redundancy. Third, the current approach deliberately avoids population-level responses so that majority views do not dominate minority ones. Whether redundancy should depend on cultural context is nonetheless a normative question, and any universal notion of political proximity is likely untenable: Switzerland (landlocked) and Indonesia (archipelagic) may reasonably disagree on whether “Boat infrastructures” and “Transportation of goods” are closely related topics. Setting this issue aside, replacing uniform weights with a behaviorally informed prior naturally leads to a metric-learning perspective, seeking a pseudo-metric $\hat{d}^{\text{beh}}$ that minimizes the expected prediction error up to negation

$$\mathcal{L}_{s,t}(\hat{d}^{\text{beh}}) = \min_{f \in \{\sigma, \text{Id}\}} \mathbb{E}_{x_i \sim \mathcal{V} \cup \mathcal{C}} [d_i(q_s, f(q_t)) - \hat{d}^{\text{beh}}(q_s, f(q_t))]^2,$$

where $d_i(q_s, f(q_t)) = |x_{i,s} - f(x_{i,t})|$ is the distance between q_s and q_t according to x_i . In this view, $\hat{d}_1^{\text{beh}}$ arises naturally, as it corresponds (up to scaling and negation) to the average distance between answers to two questions across respondents. The choice of α then becomes fundamentally normative, e.g., reflecting the fraction of the population that must agree in response patterns before two questions start sharing weight.

Such a data-driven approach raises a cold-start problem: weights are needed to generate recommendations, yet voter responses are needed to compute the distances that determine them. In practice, distances can first be computed from candidate responses and then updated online as voter data becomes available. To simulate this setting, we estimate distances from all candidates and only 5% of voters, and evaluate recommendations on the remaining voters (Figure 5): while the smallest weight decreases by up to about 25%, recommendations remain close to the uniform baseline and stable across voter subsamples, so that the behavioral defense also operates conservatively on this questionnaire.

Role of the CRW Weights. In this paper, we argue that CRW should serve as a **default alternative to uniform weighting**, that should be **replaced with voter-specific weights** whenever they are available. A natural alternative would be to apply CRW on top of personalized weights (e.g., multiplicatively with SmartVote weights $\tilde{w}_{i,t}$). The intuition is that users assign weights to questions individually, based on the perceived importance of each topic, without accounting for the global normalization effect induced by the full weight vector. As a result, duplicated questions may each be assigned high importance independently, even though this effectively dilutes the relative weight of all other questions after normalization. Such a multiplicative scheme would in principle preserve robustness to cloning by cancelling out duplicated questions—unlike the formulation considered here—but at the cost of reduced transparency in the recommendation process. Moreover, “errors” in semantic distance estimation would become effectively irreversible, making the choice of metric more critical. In practice, subtle but meaningful differences between questions may justify entirely distinct weights from informed users.

Improving Personal Weights Elicitation. The defense pipeline we propose compensates for the absence of personalized question weights for most users, addressing symptoms rather than root causes. The underlying issue lies with VAA platforms, which must better elicit voter preferences despite user fatigue. We highlight three directions: (i) introduce mandatory weight elicitation as a dedicated step (as in Wahl-O-Mat); (ii) collect coarse topic-level preferences upfront—when engagement is highest—followed by post-completion refinement at both topic and question levels; and (iii) represent weights as normalized distributions to improve interpretability, making trade-offs explicit. This could be supported by interactive visualizations (e.g., dynamic spider-web interfaces), with CRW-derived weights serving only as the default.

Limitations & Future Work. Our experiments cover two cantons, but a single VAA and election; evaluating CRW on other platforms (e.g., Wahl-O-Mat or Vote Compass) would test transferability across electoral systems and questionnaire designs, which may affect both the optimal embedding model and radius. Moreover, our clones are LLM-generated and their answers simulated, assuming respondents treat them as the original (RQ5 only partly relaxes this); measuring how voters actually answer redundant questions would require a dedicated survey. Embedding models could further be fine-tuned for this setting, e.g., by enforcing consistency across language versions of the same questionnaire, or by aligning semantic and behavioral distances through political fine-tuning [14]. Finally, several CRW design choices remain open: alternative radius distributions $\nu(r)$ could concentrate weight on empirically relevant scales, while more expressive graph constructions, such as maximal clique-based methods [5], or the original Euclidean formulation based on smooth volume overlap [4], may better handle dense questionnaires.

7 Conclusion

As VAAs increasingly influence democratic outcomes, their algorithmic integrity is a matter of critical public interest. Structural vulnerabilities in these platforms can enable subtle yet significant electoral manipulation. By introducing and validating CRW as an unsupervised mitigation strategy, this work takes a step toward safeguarding VAA integrity, ensuring that voter–candidate recommendations reflect genuine policy preferences rather than exploitation of structural vulnerabilities.

Ethical Statement

The dataset was collected and anonymized by Politools in compliance with the new Swiss Federal Act on Data Protection (nFADP), the Telecommunications Act (TCA), and applicable data protection regulations. We adhere strictly to the platform’s terms of use, in particular by not reporting results attributable to individuals. The dataset remains private, but may be shared with researchers by Politools [21] upon request.

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This appendix complements the main text as follows. Appendix A describes the Swiss electoral context and the SmartVote data, including preprocessing and the usage of importance weights. Appendix B details the methodology: clone-generation prompts, embedding-model instructions, and the selection of target questions and radii. Appendix C reports additional results in the order of the research questions: agreement between robustness measures, per-question vulnerability (RQ1), model selection (RQ2), cross-party effects of targeted attacks (RQ4), and CRW weights on the clean questionnaire under semantic and behavioral distances (RQ6). Appendix D presents complementary analyses: which questions are most vulnerable to cloning, which embedding properties predict CRW performance, an upper bound with behavioral distances, and the distance structure of the questionnaire. Finally, Appendix E lists the full 2023 questionnaire.

A Swiss Political Context & SmartVote Dataset

This section provides additional details on the Swiss electoral context, the SmartVote platform, and the dataset underlying our empirical analysis.

A.1 SmartVote and the Swiss Electoral Context

SmartVote is Switzerland’s primary Voting Advice Application, operated by the non-profit association Politools [21]. In the 2023 Swiss National Council elections, it was used by up to 20% of eligible voters [21], which makes a high-stakes platform for studying adversarial robustness in voter–candidate recommendation systems. Roughly 85% of all registered candidates completed the SmartVote questionnaire [21].

The Swiss National Council is constituted of 200 seats distributed proportionally across the 26 cantons, with each canton allocated seats roughly in proportion to its population. Elections are held every four years, and voters may only cast votes for candidates registered in their own canton [3].

Candidates standing for election are required to form *lists*, which are party- or coalition-specific slates of candidates. The number of candidates on a list corresponds to the number of seats available in the canton. Voters are presented with the option of either voting for a predefined list or composing their own custom list by selecting individual candidates across different lists. Voters may also give additional weight to a specific candidate by placing their name twice on the list, effectively granting them two votes and boosting their position within the list [3]. The total votes accumulated across all lists associated with a party determines how many seats that party wins, and seats are then assigned to candidates in order of their ranking on the list, from top to bottom. A party’s overall strength therefore depends on how frequently its candidates appear in voters’ selections, while the internal ranking on the list determines which of those candidates are ultimately elected.

The Swiss political landscape is defined by a highly stable multi-party system. Nationally, the electorate is dominated by six major parties spanning the political spectrum: the right-wing conservative Swiss People’s Party (SVP), the left-wing Social Democratic Party (SP), the centre-right FDP. The Liberals (FDP), the centrist The Centre (Centre), the left-leaning Green Party (Green), and the centrist Green Liberal Party (GLP). Together, these parties capture the vast majority of the national vote and provide the primary ideological axes along which VAAs position candidates and voters.

All experiments in this work use the SmartVote dataset for the 2023 National Council elections in the cantons of Zurich and Bern, collected and anonymized by Politools in accordance with applicable Swiss data protection law.

A.2 Questionnaire Structure

The SmartVote questionnaire is newly created for each election cycle. For the 2023 elections, it consists of 75 questions divided into three types: 60 policy questions with answer options $A_t = \{0, 25, 75, 100\}$, 7 value questions with $A_t = \{0, 17, 33, 50, 67, 83, 100\}$, and 8 budget questions with $A_t = \{0, 25, 50, 75, 100\}$. All questions accept a voter-assigned importance weight from $W_t = \{0, 0.5, 1, 2\}$, where 1 is the default for answered questions and 0 is assigned automatically to skipped questions [3].

Questions are assigned to one of 14 topic categories by SmartVote: *Federal budget*, *Energy & transport*, *Values*, *Democracy*, *Media & Digitalization*, *Nature conservation*, *Welfare state & family*, *Economy & labour*, *Foreign trade & foreign policy*, *Health*, *Security & military*, *Finances & taxes*, *Immigration & integration*, *Society & ethics*, and *Education*. The number of questions per category ranges from 3 (Education) to 8 (Federal budget); all questions are listed by category in Appendix E (Table 13).

In addition to the full 75-question survey, SmartVote offers a 30-question *rapide* version, hereafter referred to as the mini questionnaire, curated by SmartVote experts as a representative subset of the full questionnaire. The mini questionnaire covers 12 of the 14 topic categories, with the *Values* and *Federal budget* questions only occurring in the full questionnaire (see the *Mini* column of Table 13).

A.3 Data Cleaning

The raw dataset was preprocessed following the procedure described by Berdoz et al. [3]: corrupt entries lacking essential metadata were removed, voters who answered fewer than 15 questions were excluded ($< 1.3\%$ of the dataset), and duplicate entries were resolved by retaining only the most recent recommendation with the highest number of answered questions per unique voter ID. Recommendations submitted before the questionnaire went live or after the election date were filtered out, as were responses exhibiting random clicking behaviour (more than 14 consecutive identical answers, $\approx 0.1\%$). On the candidate side, candidates who did not participate in SmartVote were removed, and youth party

affiliations were merged with their parent parties (e.g., JUSO with SP, JSVP with SVP).

After preprocessing, the Zurich dataset contains 41,658 voters and 1,029 candidates, and the Bern dataset 34,052 voters and 685 candidates; SmartVote displays the top 36 and 24 candidates, respectively, matching each canton’s number of National Council seats.

A.4 Usage of Importance Weights

In the whole 2023 Swiss National Council elections dataset, only 45.8% of all voters (86,138) entered at least one value different from the default weight of one. Among voters that modified at least one value, the average number of importance weights different from one is 11.35, representing only 15.1% of all questions.

B Additional Methodological Details

This section provides additional methodological details underlying the experimental setup, including clone generation, embedding model instructions, and procedures for model and hyperparameter selection.

B.1 Clone Generation: Prompts and Examples

All synthetic clones are generated using GPT-4o (gpt-4o-2024-08-06) with temperature 0.8 and structured JSON output. Each prompt follows a shared template:

```
You are helping generate test data for a Swiss political
questionnaire.

Original question:
"{question_text}"

Task: {type-specific instruction}

Return a JSON object with a single key "paraphrase" containing
your result.
Example: {"paraphrase": "Your paraphrase here."}
```

When multiple clones of the same type are needed for a single question, all previously generated paraphrases of that type are appended to the prompt with the instruction: “The following paraphrases of this type have already been generated. You MUST produce a DIFFERENT one.” For the hard paraphrase and hard negation types, up to three previously generated easy paraphrases are additionally included as negative exemplars with the prefix: “For reference, these are EASY paraphrases (similar wording). Your hard paraphrase must be MUCH more different than these,” followed by the supplementary rules shown below.

Easy Paraphrase

Rephrase the question using similar but not identical wording and a slightly different sentence structure. The meaning must be exactly the same. Some words may overlap with the original.

Hard Paraphrase

Rephrase the question using COMPLETELY DIFFERENT words and sentence structure. The meaning must be identical, but you MUST NOT share any significant content words with the original. Use synonyms, different grammatical forms, or circumlocutions. You may change the grammatical form (e.g. question to statement or vice versa).

For example: if the original says 'taxes', use 'fiscal levies' or 'government revenue'. If it says 'legalized', use 'permitted by law' or 'decriminalized'. If it says 'retirement age', use 'the point at which workers leave the workforce' or 'the threshold for ending one's professional career'.

IMPORTANT RULES:

- You MUST NOT reuse any significant content words from the original question.
- Nouns, verbs, and adjectives that carry the topic meaning must ALL be replaced with synonyms or circumlocutions.
- If you find yourself writing a word that appears in the original, STOP and find an alternative.
- The result should look like it was written by someone who never saw the original, describing the same concept from scratch.

Easy Negation

First, create an easy paraphrase of the original (similar wording, slightly different structure). Then negate that paraphrase so that someone who agrees with the original would disagree with your version. Return only the final negated paraphrase.

Hard Negation

First, create a hard paraphrase of the original -- using COMPLETELY DIFFERENT vocabulary and sentence structure, with NO significant content words from the original. Then negate that hard paraphrase so that someone who agrees with the original would disagree with your version. Return only the final negated paraphrase.

For example: if the original says 'taxes', the hard paraphrase should use 'fiscal levies' or 'government revenue' instead. Then negate that rephrased version.

IMPORTANT RULES:

- You MUST NOT reuse any significant content words from the original question.

- Nouns, verbs, and adjectives that carry the topic meaning must ALL be replaced with synonyms or circumlocutions.
- If you find yourself writing a word that appears in the original, STOP and find an alternative.
- The result should look like it was written by someone who never saw the original, describing the same concept from scratch.

The generated clones were manually inspected to verify that they address the same policy issue as the original question. Table 2 finally shows an example of all four clone types for a representative question.

Condition	Question text
Original	Should automatic facial recognition be banned in public spaces?
Easy paraphrase	Should the use of automatic facial recognition in public locations be prohibited?
Hard paraphrase	Is it appropriate to forbid the deployment of systems that identify people by their faces in open-access environments?
Negation (easy)	Should we permit the use of automatic facial recognition in public spaces?
Negation (hard)	Is it advisable to allow the deployment of systems that identify people by their faces in shared community areas?

Table 2: Example clones for Q32267, the second most impactful question at $\ell = 4$. The mixed setting combines one clone of each of the four types.

B.2 Details regarding Model and α Selection

This section covers additional methodological details regarding model and α selection left out of the main body for lack of space.

Embedding Model Instructions Four models accept task-specific instructions prepended to the input text. The instructions used are shown in Table 3. Models with no prefix receive raw question text. Instruction-aware models receive either a prefix template or a task keyword passed as an encode argument.

To ensure a consistent distance metric for CRW when using an asymmetric approach, the distance between any two questions q_i and q_j is symmetrized by averaging the two embedding distances:

$$d^*(q_i, q_j) = \frac{d(q_i, q_j) + d(q_j, q_i)}{2}$$

where $d(q_i, q_j)$ represents the distance from q_i to q_j under the model’s specific instruction set.

Model	Input passed to the encoder
SBERT, BGE-M3, GTE	raw question text
E5	<code>query: {text}</code>
E5-Asym	<code>query: {text}</code> and <code>passage: {text}</code> (asymmetric)
Nomic v2	<code>clustering: {text}</code>
Jina v3	raw text, with <code>embedding_task="separation"</code>
E5-Instruct	<code>Instruction: {instr₁}\nQuery: {text}</code>
E5-Asym-Instruct	as E5-Instruct, encoded asymmetrically
Qwen3	raw text, with <code>prompt={instr₂}</code>

`instr1`: *Identify the political topic discussed in the following question, regardless of the stance or position taken.*

`instr2`: *Classify the following political question by its topic or policy area, ignoring whether the question is for or against the policy.*

Table 3: Input format of each embedding model. Instruction-tuned models receive the task instruction given below the table; `embedding_task` and `prompt` are arguments of the model’s encoding function.

Most Impactful Question Selection To identify the most impactful questions, we craft for each question $q \in \mathcal{Q}$ a manipulated questionnaire $\mathcal{Q}' = \mathcal{Q} \cup \{q'_1, \dots, q'_4\}$, where each q'_s is a 0-behavioral 0-embedding clone of q .

Let τ denote Kendall’s rank correlation, defined by

$$\tau(r_i, r'_i) = \frac{|\mathcal{C}^+| - |\mathcal{C}^-|}{\binom{C}{2}}, \quad (1)$$

where $\mathcal{C}^+$ and $\mathcal{C}^-$ denote the sets of concordant and discordant candidate pairs, i.e., pairs $\{c_j, c_{j'}\}$ ordered identically or oppositely by r_i and r'_i , and C is the number of candidates.

For every robustness measure $m \in \{J_{36}, \rho_S, \tau\}$, we then compute the average recommendation robustness across voters, and rank questions by the robustness of their corresponding attack set for each of the metric individually. We select the five questions with the lowest mean rank across all three metrics. Table 4 displays the resulting five most impactful questions.

*Selection of optimal radii α^** To select the best radius α^* for each model, we compute for each metric $m \in \{J_{36}, \rho_S, \tau\}$ the average recommendation robustness across all voters and attack settings. We then aggregate these three measures of robustness within a composite score that ensures scale invariance across measures, i.e.,

$$\text{Score}(\text{model}, \alpha) = \sum_{m \in \{J_{36}, \rho_S, \tau\}} \frac{m_{\text{model}}^{\text{CRW}}(\alpha) - \min m_{\text{model}}^{\text{CRW}}}{\max m_{\text{model}}^{\text{CRW}} - \min m_{\text{model}}^{\text{CRW}}}, \quad (2)$$

Rank	Q ID	Question	Category
1	32214	Do you support an increase in the retirement age (e.g., to 67)?	Welfare state & family
2	32234	Should a third official gender be introduced alongside “female” and “male”?	Society & ethics
3	32240	Are you in favor of introducing a minimum wage of CHF 4,000 for all full-time employees?	Economy & labour
4	32261	A popular initiative aims to reduce television and radio fees (CHF 200 per household, exemption for businesses). Do you support this initiative?	Democracy, Media & Digitalization
5	32228	Should the conditions for naturalization be relaxed (e.g., shorter residence period)?	Immigration & integration

Table 4: The five most impactful questions, i.e., those with the lowest mean rank across J_{36} , ρ_S , and τ after injecting $\ell = 4$ exact copies under uniform weights (Q32240 and Q32261 tie for third). All five happen to be included in the mini questionnaire.

where $m_{\text{model}}^{\text{CRW}}(\alpha)$ denotes the average robustness across voters and attacks according to measure m at radius α , and the min and max are taken over all values of α , for a fixed model.

For each model, we then select the value α^* that maximizes the function $\text{Score}(\text{model}, \cdot)$.

B.3 Question Selection for Targeted Party Attacks

For each of the six major parties, the five target questions of RQ4 are selected by screening the full questionnaire. Each of the 75 questions is duplicated individually with four exact copies, and the resulting change in top-36 visibility $\Delta\nu_{36}(p)$ is recorded for each party p under uniform weights. For each party, we then retain the five questions with the largest gain $\Delta\nu_{36}(p)$, listed in Table 5. Since this screening uses the full voter response data, the selection is only available to an omniscient adversary. Several questions benefit more than one party, such as the minimum wage (SP and Green) or asylum centers and nuclear energy (FDP and SVP).

C Additional Results

This section provides additional experimental results, covering robustness measures agreement, per-question distortion analyses, model and hyperparameter selection, and robustness checks on the full questionnaire.

Target Selected questions	
SP	Compulsory vaccination (Q32220), Relaxed environmental rules for renewables (Q32248), Minimum wage (Q32240), Occupational pension cuts (Q32217), US free-trade agreement (Q32270)
Green	Occupational pension cuts (Q32217), Minimum wage (Q32240), Progressive electricity tariffs (Q32251), Sugar tax (Q32221), Arms re-export (Q32266)
GLP	Health-insurance deductible (Q32222), Electricity-market liberalization (Q32242), Active euthanasia (Q32233), Retirement age (Q32214), Individual taxation (Q32237)
Centre	Highway widening (Q32245), Naturalization (Q32228), Voting age 16 (Q32262), Federal tax cuts (Q32236), Fiscal equalization (Q32239)
FDP	Asylum centers outside Europe (Q32230), New nuclear plants (Q32249), TV and radio fees (Q32261), Corporate supply-chain responsibility (Q32271), Equal-pay controls (Q32244)
SVP	Asylum centers outside Europe (Q32230), Equal educational opportunities (Q32227), Active euthanasia (Q32233), Third gender (Q32234), New nuclear plants (Q32249)

Table 5: Questions selected for the targeted attack on each party, i.e., the five questions whose duplication most increases that party’s top-36 visibility under uniform weights.

C.1 Agreement between Robustness Measures

To justify reporting Jaccard similarity as the sole robustness measure throughout the main text, we verify that all three ranking measures $m \in \{J_{36}, \rho_S, \tau\}$, cf. Equation (1), agree on the relative severity of distortion across questions and clone intensities. For each of the 75 questions cloned at 10 intensities ($\ell \in \{1, 2, 3, 4, 5, 7, 10, 15, 20, 30\}$) with identical clones under uniform weights, we compute the mean distortion across all voters under each metric.

Figure 6 shows the pairwise relationships across all 750 (question, ℓ) combinations. The three measure pairs exhibit near-perfect rank agreement: Spearman correlations range from $\rho_S = 0.965$ (Jaccard vs. Spearman’s ρ_S) to $\rho_S = 0.994$ (Spearman’s ρ_S vs. Kendall’s τ). The metrics rank questions in effectively the same order at every clone intensity, and the agreement holds across all subsequent experiments—the metric used does not affect which model is selected, which α is optimal, or which questions are identified as most vulnerable.

Given this strong agreement, we adopt Jaccard as the primary measure for two reasons. First, it is the most directly relevant to electoral outcomes. SmartVote displays the top 36 candidates to each voter, corresponding to the number of National Council seats in Canton Zurich. Voters who fully adopt this recommended list determine the party seat allocation; Jaccard over the top-36 set therefore measures exactly what matters—whether the same candidates appear in a voter’s recommendation, regardless of their internal ordering. Spearman’s ρ_S and Kendall’s τ , by contrast, are sensitive to rank permutations deep in the list that have no practical consequence for voter behavior or party visibility. Sec-

ond, Jaccard has a wider dynamic range on the distortions we study: at $\ell = 4$ identical clones, Jaccard values span 0.657–0.928 across questions, while Spearman’s ρ_S compresses into 0.92–0.99 and Kendall’s τ into 0.80–0.97. The greater spread makes Jaccard more informative for discriminating between models, α values, and attack settings.

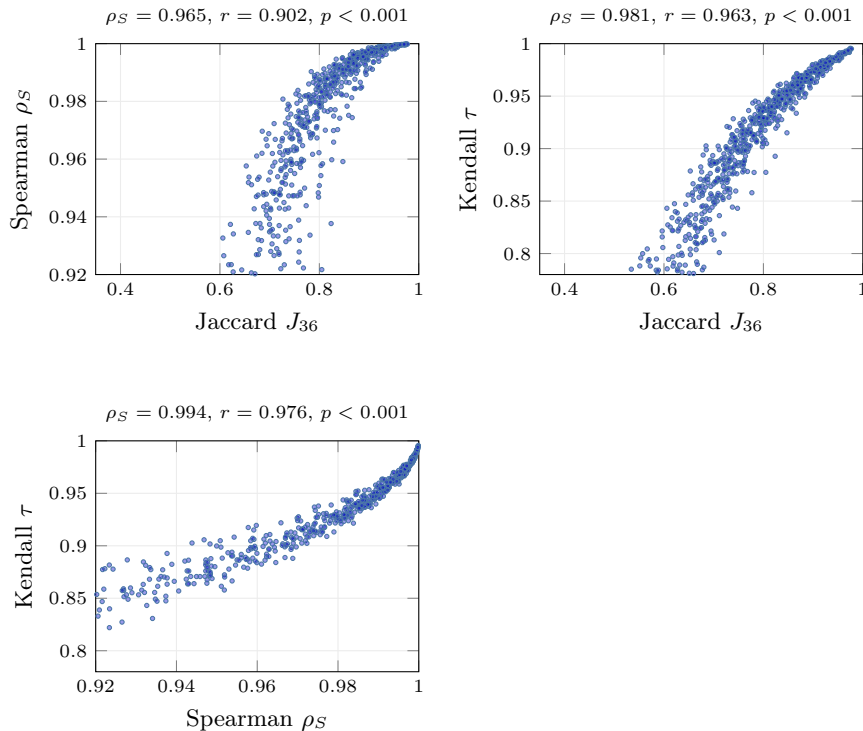


Fig. 6: Pairwise agreement between Jaccard similarity (J_{36}), Spearman’s ρ_S , and Kendall’s τ across all 75 questions and 10 clone counts ($\ell \in \{1, \dots, 30\}$, $n = 750$ points per panel). Each dot represents one (question, ℓ) combination under identical cloning with uniform weights. All three metric pairs show near-perfect rank agreement ($\rho_S > 0.965$, $p < 0.001$), justifying the use of Jaccard as the primary metric. Spearman and Kendall are the most tightly coupled ($\rho_S = 0.994$), while Jaccard shows slightly more spread relative to both rank-based metrics due to its sensitivity to the top-36 boundary.

C.2 Vulnerability Assessment

Figure 7 tracks the evolution of J_{36} as ℓ exact copies are injected, for the questions with maximal (Q32214), median (Q32242), and minimal (Q32235) distor-

tion at $\ell = 30$. Distortion appears immediately: across all 75 questions, a single copy already reduces J_{36} to between 0.847 and 0.977. It then grows monotonically with ℓ for every question, with diminishing marginal effect per added copy, consistent with the effective weight $\frac{1+\ell}{Q+\ell}$ of the targeted question saturating as ℓ grows. At $\ell = 30$, the most vulnerable question retains only $J_{36} = 0.388$, i.e., about 16 of each voter’s top-36 candidates are replaced on average. Spearman’s ρ_S evaluated over the full candidate list exhibits a similar monotonic decay.

Figure 8 reports the distortion $1 - J_{36}$ obtained at $\ell = 4$ for all 75 questions. Every question induces measurable distortion, ranging from 0.072 (federal spending on public security, Q32286) to 0.343 (retirement age, Q32214). This confirms that redundancy is a structural vulnerability of the questionnaire rather than an artifact of a few polarizing topics. Distortion is however highly heterogeneous across questions; we investigate its predictors in Appendix D.1, finding that answer variance and skip rate are the dominant factors.

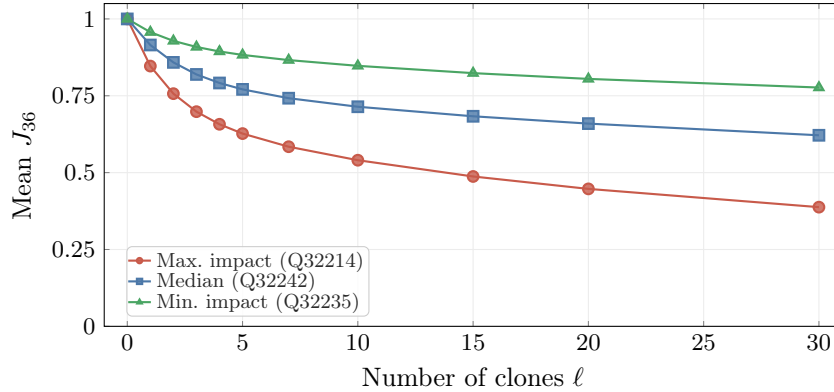


Fig. 7: Mean Jaccard similarity (top-36) as a function of the number ℓ of exact copies injected, for the questions with maximal, median, and minimal distortion at $\ell = 30$, under uniform weights. Distortion appears with a single copy and grows monotonically with ℓ ; Spearman’s ρ_S over the full candidate list ($C = 1,029$) exhibits a similar decay.

C.3 Model and α Selection

We present and discuss in this section a few additional results regarding the model selection step.

Clone-Type Variation. Figure 9 displays the rank of each model within each clone condition, ordered by difficulty. The ranking reveals E5-Instruct’s broad dominance: it ranks first across four of the five conditions, encompassing both paraphrases and negations at both difficulty levels. The notable exception is the

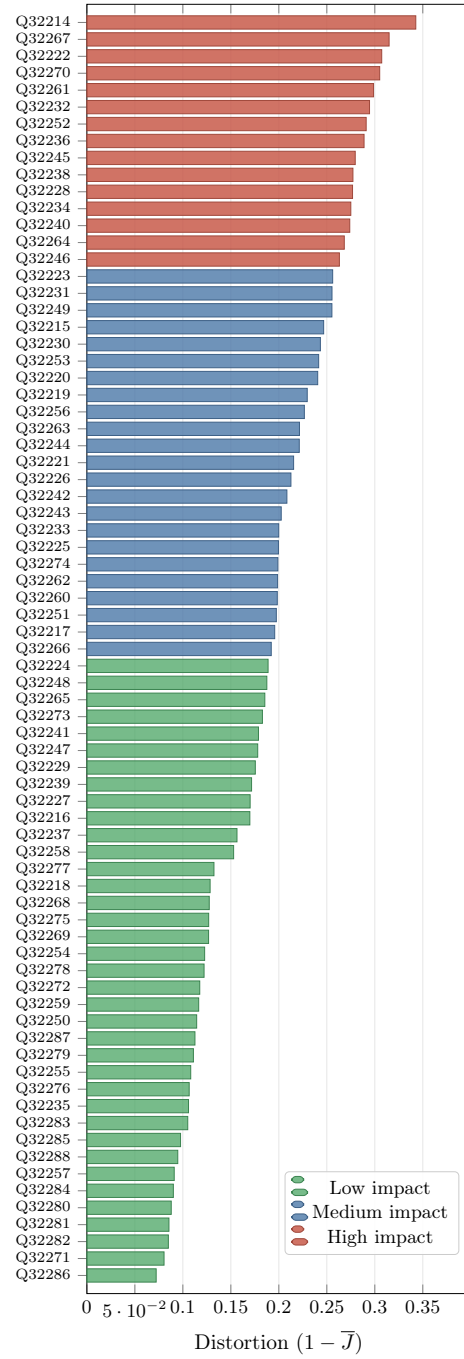


Fig. 8: Per-question distortion $1 - J_{36}$ (mean over voters) after injecting $\ell = 4$ exact copies of each question under uniform weights. Questions are ordered by distortion; colors mark the bottom 50%, the 50th–80th percentile, and the top 20% of questions.

mixed setting, where E5-Instruct drops to third, outperformed by Jina-v3 and Qwen3. This presents a curious anomaly: a model that excels at detecting individual clone strategies in isolation struggles relatively when those strategies are deployed simultaneously. Beneath the top tier, there is significant rank volatility. Qwen3, for instance, overtakes GTE and BGE-M3 on easy negations, but plummets to 8th on hard negations, where GTE and BGE-M3 recover their standings. Finally, E5-Asym-Instruct remains anchored at the bottom, ranking 10th on four of the five conditions. It escapes last place only on hard negations, where 3 other models perform even worse.

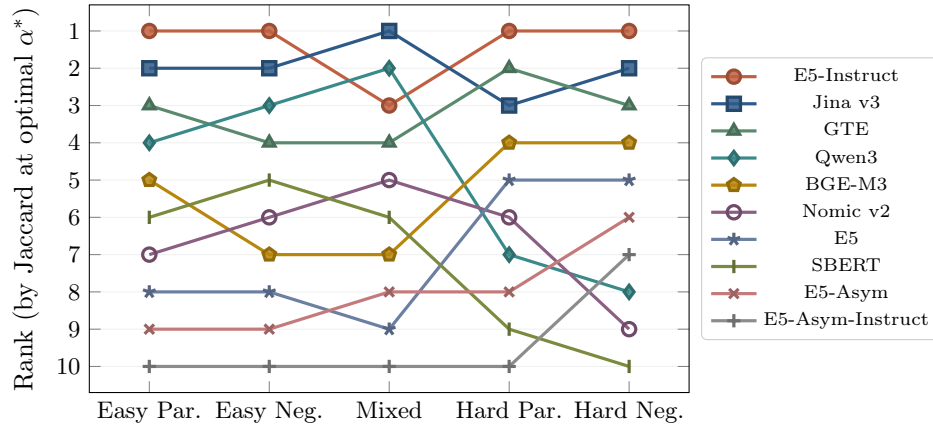


Fig. 9: Model ranking by clone type, based on maximum Jaccard similarity at each model’s optimal α^* . Attack settings are ordered by approximate difficulty (left to right). The top two models (E5-Instruct, Jina v3) maintain near-constant rankings across all conditions, while mid-tier models exhibit considerable reshuffling: Qwen3 ranks 2nd on the mixed setting but drops to 8th on hard negations, and E5-Asym-Instruct ranks last on four conditions but climbs to 7th on hard negations.

Selected Configuration. Driven by the composite ranking procedure, the selected defense configuration is **E5-Instruct** paired with a CRW radius of $\alpha^* = 0.4$. While E5-Instruct’s performance actually peaks at $\alpha = 0.3$ under certain conditions (most notably the easy paraphrase condition) expanding the radius to 0.4 maximizes the aggregate composite score across the full spectrum of difficulty tiers. The rationale for favoring this slightly larger radius is rooted in the threat model: a strict threshold of $\alpha = 0.3$ operates too conservatively, successfully capturing obvious lexical duplicates but risking the exclusion of more complex, structurally distinct clones (such as hard paraphrases and hard negations) that naturally sit at slightly higher embedding distances. Furthermore, because our parameter sweep evaluated discrete intervals, the “true” continuous

optimum likely resides somewhere between 0.3 and 0.4. However, as this framework serves primarily as a proof of concept, pinpointing a hyper-exact scalar value is less critical than establishing a stable, robust threshold capable of generalizing across diverse attack vectors. This configuration is therefore utilized in all subsequent experiments unless otherwise stated. The underlying structural mechanisms driving this performance—and specifically why E5-Instruct outperforms models that more closely mimic ground-truth correlation metrics—are dissected in Appendix D.

C.4 Generalization on Zurich

Figure 10 repeats the RQ3 evaluation on Zurich, the canton used for selection. Since RQ2 only involved five target questions, the remaining 70 questions are unseen there as well. CRW improves robustness in 370 of 375 question–attack pairs (98.7%), the remaining five showing negligible degradation (< 0.001), and the mitigated share follows the same hierarchy as on Bern, from 47% for easy paraphrases to 17% for hard negations. Results are consistent across cantons: per-question improvements on Zurich and Bern are almost perfectly rank-correlated (Spearman $\rho_S = 0.99$), indicating that the defense’s effect is driven by the questions and clones rather than by the electorate.

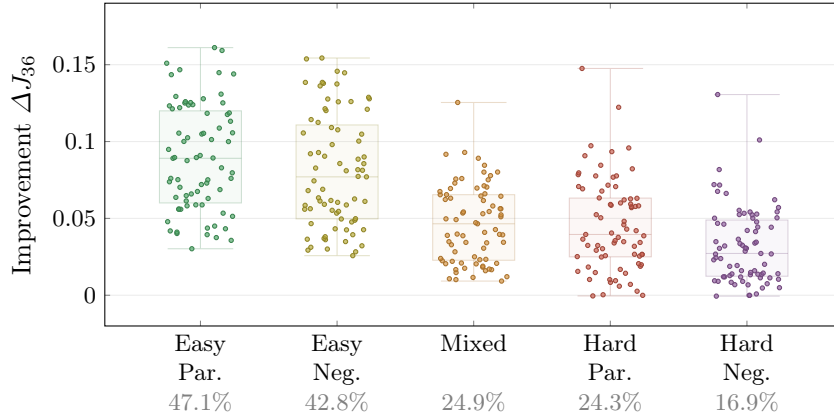


Fig. 10: Per-question CRW improvement ΔJ_{36} on Zurich across 75 questions and 5 attack settings (E5-Instruct, $\alpha^* = 0.4$). Gray percentages indicate the mean share of distortion mitigated $\overline{\Delta J_{36}}/(1 - J_{36})$, with uniform baseline $J_{36} = 0.811$ across all conditions.

C.5 Targeted Party Manipulation

Table 6 reports the full cross-party impact of the high-effort attacks on Bern under uniform weights, complementing Table 1 in the main text. Without CRW,

the targeted party gains between +0.82pp (SVP) and +5.07pp (SP) under the high-effort attack, and between +0.26pp (SVP) and +2.16pp (SP) under the low-effort attack; in both cases, SP benefits most, followed by GLP.

Since visibility is zero-sum, these gains come at the expense of other parties, but the collateral impact is not evenly distributed across the political spectrum. When the Centre or left-leaning parties are targeted, they largely draw visibility from one another, while FDP and SVP remain relatively insulated: the high-effort attack favoring the Centre costs SP -3.09 pp and the Green -2.20 pp, and the one favoring the Green costs GLP -3.26 pp, while FDP and SVP change by at most 0.70pp in all four cases. Conversely, attacks favoring FDP or SVP mostly draw from SP and benefit neighboring parties: the attack favoring the SVP raises the Centre’s visibility (+1.93pp) more than the SVP’s own (+0.82pp).

CRW mitigates these collateral effects alongside the targeted gain. Under the high-effort attack favoring the Centre, the losses of SP and the Green shrink from -3.09 and -2.20 pp to -2.46 and -1.64 pp, respectively.

Attack	Visibility Change (pp)					
	Centre	SP	GLP	Green	FDP	SVP
Centre	+3.44	-3.09	+0.13	-2.20	+0.70	+0.15
SP	-1.10	+5.07	-2.84	-0.61	-0.11	-0.33
GLP	-0.21	-0.82	+4.18	-0.67	+0.14	+0.02
Green	-1.48	+1.65	-3.26	+3.26	-0.44	-0.26
FDP	+0.08	-1.00	+0.41	-0.57	+1.03	+0.13
SVP	+1.93	-0.64	-0.04	-0.27	+0.63	+0.82

Table 6: Cross-party visibility impact of high-effort attacks on Bern (one clone of each type per target question, 20 clones in total). Each row corresponds to an attack designed to benefit the row’s party, while columns show the resulting visibility change (in percentage points) for all six major parties under uniform weights. Diagonal entries (bold) are the targeted party’s gain. Off-diagonal entries reveal collateral effects: an attack benefiting one party systematically harms others, though the magnitude and distribution of harm varies across the political spectrum.

For comparison, Tables 7 and 8 report the same experiment on Zurich. Results are closely consistent with Bern: mean gains of +1.23pp and +3.08pp under low- and high-effort attacks (vs. +1.18pp and +2.97pp on Bern), of which CRW mitigates on average 56% and 27% (55% and 28% on Bern). The main difference lies in which party benefits most (the Centre on Zurich, SP on Bern), while the collateral pattern is similar: attacks favoring the Centre or left-leaning parties draw visibility mostly from one another, leaving FDP and SVP largely unaffected.

Low Effort ($\ell = 1$) High Effort ($\ell = 4$)							Attack						
Party	$\Delta\nu_{36}$			$\Delta\nu_{36}^{\text{CRW}}$			Benefits	Visibility Change (pp)					
	$\Delta\nu_{36}$	$\Delta\nu_{36}^{\text{CRW}}$	Mitig.	$\Delta\nu_{36}$	$\Delta\nu_{36}^{\text{CRW}}$	Mitig.		Centre	SP	GLP	Green	FDP	SVP
Centre	+1.83	+1.05	42.6%	+4.76	+3.49	26.7%	Centre	+4.76	-2.35	-1.33	-2.39	+0.19	+0.32
SP	+1.76	+0.83	53.0%	+4.53	+3.37	25.7%	SP	-1.20	+4.53	-1.85	-0.57	-0.51	-0.26
GLP	+1.74	+0.67	61.8%	+4.16	+2.92	29.8%	GLP	-0.77	-1.40	+4.16	+0.64	+0.17	+0.17
Green	+0.99	+0.41	58.8%	+2.20	+1.77	19.4%	Green	-1.45	+1.19	-1.63	+2.20	-0.58	-0.37
FDP	+0.58	+0.24	58.1%	+1.44	+1.05	26.9%	FDP	-0.88	-1.23	-0.83	-0.54	+1.44	+0.71
SVP	+0.47	+0.19	60.5%	+1.37	+0.93	31.7%	SVP	-0.84	-0.17	+0.23	-0.63	+1.27	+1.37
Mean	+1.23	+0.56	55.8%	+3.08	+2.26	26.7%							

Table 7: Visibility gain $\Delta\nu_{36}$ (pp) of each targeted party on Zurich relative to the unmanipulated questionnaire, under uniform weights and CRW (E5-Instruct, $\alpha^* = 0.4$), with $n = 5$ target questions per party. “Mitig.” gives the share of the gain mitigated by CRW.

Table 8: Cross-party visibility change (pp) on Zurich under high-effort attacks and uniform weights. Each row is an attack favoring the row’s party, each column the resulting change for one party; diagonal entries (bold) give the targeted party’s gain.

C.6 Impact on Clean Questionnaire

Figure 11 shows the CRW weight of each question on the unmanipulated questionnaire, computed with the configuration selected in RQ2. Weights remain close to uniform throughout: 55 of the 75 questions gain slightly in weight, with 33 of them reaching the maximum of +3.3% above uniform, which corresponds to questions with no close neighbor in embedding space. The remaining 20 questions lose weight, by at most 16%. These reductions concentrate on a few clusters of topically related questions: electricity-market liberalization (Q32242) and progressive electricity tariffs (Q32251), both at −16%, health-insurance deductibles (Q32222, −14%) and premium subsidies (Q32215, −11%), as well as the pension questions of the Welfare block (Q32216–Q32218, −7 to −9%) and the transport-related budget items (Q32284–Q32285, −9%). Such pairs measure related but distinct policy dimensions, so CRW’s mild down-weighting reflects partial rather than full redundancy, and, as shown in RQ6, has little effect on recommendations.

For comparison, Figure 12 shows the CRW weights obtained with the behavioral distance d_1^{beh} , i.e., the mean absolute answer difference up to negation, divided by 100. We show the configuration with the largest weight reduction in Figure 5, namely $\alpha = 0.205$ on voter subsample $s = 2$: if disagreements were full reversals, two questions start sharing weight only when fewer than about 20% of respondents answer them differently. Even then, weights remain close to uniform: 61 questions lie slightly above it (at most +1.4%), and voters retain on average $J_{36} = 0.963$ of their top-36 candidates. Sharing is concentrated on Schengen and the Bilateral Agreements (Q68/Q72, −25% each), which have the closest response patterns in the questionnaire ($d_1^{\text{beh}} \approx 0.10$), and on the education and public-transport budgets (Q82/Q84, −8%); nine further questions lose at most 3%. Unlike E5-Instruct, which mostly discounts pairs on the same topic,

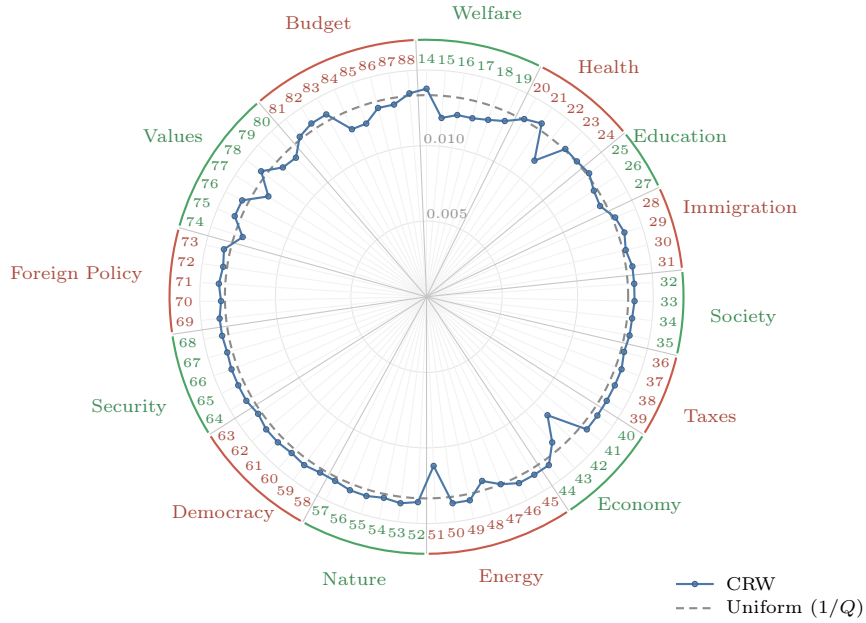


Fig. 11: Normalized CRW weight of each SmartVote question on the unmanipulated questionnaire (E5-Instruct, $\alpha^* = 0.4$), compared with the uniform weight $1/Q \approx 0.0133$ (dashed circle). Questions are grouped by topic; labels give the last two digits of the question ID (Q322xx). Weights range from 0.0112 to 0.0138, the largest reductions affecting pairs of topically overlapping questions.

d_1^{beh} also merges topically distinct questions answered along the same left-right divide, as anticipated in Section 3.

D Complementary Analyses

This section presents four complementary analyses. Section D.1 investigates which factors best explain why certain questions are more potent vehicles for redundancy attacks. Section D.2 examines the structural properties of sentence embedding models that determine their effectiveness when combined with CRW. Section D.3 then establishes an upper bound on CRW performance using a behavioral distance, separating the limitations of the embedding from those of the framework itself. Finally, Section D.4 studies the global geometry of the SmartVote dataset under both embedding and behavioral correlation metrics.

D.1 Predictors of Question Duplication Impact

Figure 8 shows that clone impact varies widely across questions, with distortion $1 - \bar{J}_{36}$ ranging from 0.072 to 0.343 at $\ell = 4$. Table 9 reports the Spearman

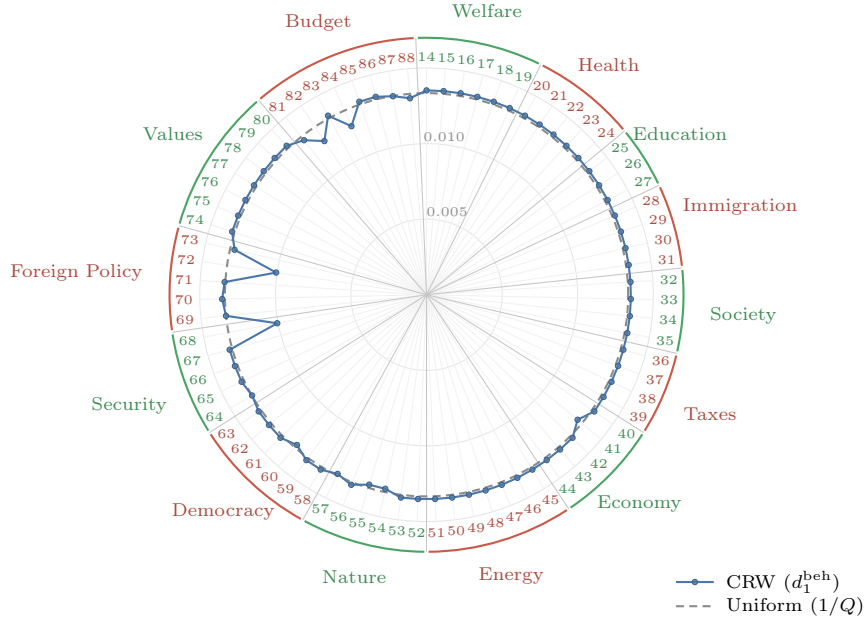


Fig. 12: Normalized CRW weights on the unmanipulated questionnaire under the behavioral distance d_1^{beh} (Zurich, $\alpha = 0.205$, voter subsample $s = 2$), i.e., the configuration with the largest weight reduction in Figure 5; same layout as Figure 11. Weights range from 0.0100 to 0.0135, sharing being concentrated on Q68/Q72 and Q82/Q84.

correlation of question-level properties with this impact, and with the impact on party visibility ($\max_p |\Delta \nu_{36}(p)|$).

Answer variance is the dominant predictor ($\rho_S = +0.81$ for voter variance): cloning a question multiplies its contribution to the squared distance, which matters most when respondents spread across the answer scale. *Skip rate* is a strong negative predictor ($\rho_S = -0.65$), since skipped questions receive weight zero; question position acts as a proxy, as later questions are skipped more often [3]. At the party level, the product of voter and candidate variance ranks first: a question on which candidates agree cannot shift visibility between parties, however polarized voters are.

In contrast, *inter-question correlation* barely predicts impact. One might expect cloning a question already correlated with others to matter less, as its policy dimension is already weighted. All eight correlation-based measures are indeed slightly negative, but only one reaches significance (mean voter r^2 , at the party level), and with $n = 75$ questions only effects of $|\rho_S| \gtrsim 0.23$ are detectable. This is consistent with the low redundancy of the SmartVote questionnaire (mean $|\rho| = 0.19$, maximum 0.78; Appendix D.4), which leaves answer variance as the main driver of vulnerability.

Predictor	Recommendations Party visibility			
	ρ_S	p	ρ_S	p
Voter answer variance	+0.806	< 0.001	+0.604	< 0.001
Combined variance (voter $\times$ cand.)	+0.782	< 0.001	+0.621	< 0.001
Candidate answer variance	+0.675	< 0.001	+0.598	< 0.001
Voter skip rate	-0.654	< 0.001	-0.358	0.002
Question position in survey	-0.556	< 0.001	-0.576	< 0.001
Max $ \rho $ with others (voter)	-0.108	0.357	-0.182	0.119
Max $ \rho $ with others (candidate)	-0.084	0.471	-0.152	0.193
Mean r^2 with others (voter)	-0.148	0.205	-0.244	0.035
Mean r^2 with others (candidate)	-0.086	0.463	-0.123	0.291
Sum r^2 with others (voter)	-0.157	0.179	-0.203	0.080
Sum r^2 with others (candidate)	-0.040	0.735	-0.096	0.413
Correlated neighbours (voter)	-0.161	0.169	-0.168	0.149
Correlated neighbours (candidate)	-0.051	0.662	-0.126	0.282

Table 9: Spearman correlation between question properties and the impact of $\ell = 4$ exact copies under uniform weights, on recommendations ($1 - \bar{J}_{36}$) and on party visibility ($\max_p |\Delta\nu_{36}(p)|$), across the 75 questions of Zurich. Gray: $p \geq 0.05$.

D.2 Predictors of Embedding Model Performances

The model selection step in Section 5 identifies E5-Instruct as the best-performing model, but does not explain *why*. To understand which embedding properties drive CRW effectiveness, we construct a diagnostic benchmark from the 2019 SmartVote questionnaire (an independent dataset from the 2023 questionnaire used in all other experiments). We randomly select 14 questions, and for each of these anchor questions, we generate same-topic paraphrases and syntactic/topical trap questions using GPT-4o.² All generated pairs were manually inspected for quality.

To formalize these metrics, let q^* denote an anchor question. Let $\mathcal{S}$ be the set of same-topic variants (both paraphrases and negations), $\mathcal{N} \subset \mathcal{S}$ be the subset of strictly negated variants, and $\mathcal{T}$ be the set of trap variants. Let $d(x, y)$ represent the spatial embedding distance between two questions. For each anchor, the benchmark calculates the following three metrics (which are then averaged across all anchors):

- **Topic coherence (lower is better)**: the ratio of the mean distance to same-topic variants over the mean distance to trap variants. Rather than measuring raw distance, this ratio captures whether a model successfully clusters functionally related questions tighter than deceptive, keyword-sharing

² (gpt-4o-2024-08-06) with temperature 0.8 and structured JSON output.

traps. Formally:

$$\text{Topic Coherence}(q^*) = \frac{\frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} d(q^*, s)}{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d(q^*, t)}$$

- **Negation invariance (lower is better)**: the ratio of the mean distance to negated variants over the mean distance to trap variants. For CRW purposes, a question and its exact policy opposite (e.g., “Should immigration be increased?” and “Should immigration be restricted?”) address the identical functional dimension. The model must place them closer together than a keyword trap, despite being semantically opposite. Formally:

$$\text{Negation Invariance}(q^*) = \frac{\frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} d(q^*, n)}{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d(q^*, t)}$$

- **Ordering accuracy (higher is better)**: the percentage of all possible combinations of (same-topic variant, trap variant) where the model correctly places the same-topic variant closer to the anchor than the trap. A score of 100% indicates perfect spatial separation between the valid policy dimension and the deceptive surface-level text. Formally:

$$\text{Ordering Accuracy}(q^*) = \frac{1}{|\mathcal{S}||\mathcal{T}|} \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}} \mathbb{I}(d(q^*, s) < d(q^*, t))$$

where $\mathbb{I}$ is the indicator function returning 1 if the distance condition is met and 0 otherwise.

General Benchmark Performance. As detailed in Table 10, the evaluated models exhibit stark differences in their ability to handle topical redundancy versus surface-level deception. The top-tier models, E5-Instruct and Jina v3, demonstrate exceptional topic coherence (0.368 and 0.409, respectively), indicating they successfully compress same-topic variants to roughly a third of the distance of deceptive keyword traps. Jina v3 is particularly robust to negations, achieving a near-perfect negation invariance score of 0.109. Conversely, models at the bottom of the ranking, such as E5-Asym and E5-Asym-Instruct, struggle significantly with semantic traps: their topic coherence scores approach 1.0 (0.808 and 0.883), meaning they place entirely unrelated questions sharing surface-level vocabulary almost as close together as genuine policy paraphrases.

Interestingly, global ordering accuracy remains relatively high across the board, with seven of the ten models exceeding 90%. E5-Asym-Instruct perfectly illustrates a critical disconnect here: despite its abysmal topic coherence, it still achieves 94.0% ordering accuracy.

Topic coherence dominates. Table 10 reports the performance of each model for each benchmark metric, while Table 11 reports the Spearman correlation between each benchmark metric and CRW performance $J_{36}^{\max}$ across the 10

Model	CRW $J_{36}^{\max}$	Predict CRW		Do not predict CRW	
		Topic coh. ↓	Neg. inv. ↓	Ord. acc. ↑	Corr. ρ_S
E5-Instruct	0.585	0.368	0.198	99.4%	+0.105
Jina v3	0.581	0.409	0.109	99.7%	+0.192
GTE	0.563	0.604	0.247	90.2%	+0.166
Qwen3	0.565	0.459	0.146	98.5%	+0.198
BGE-M3	0.552	0.674	0.320	88.1%	+0.112
Nomic v2	0.542	0.631	0.220	93.2%	+0.128
E5	0.530	0.722	0.336	80.4%	+0.068
SBERT	0.541	0.587	0.134	92.3%	+0.060
E5-Asym	0.518	0.808	0.637	85.1%	+0.075
E5-Asym-Instruct	0.491	0.883	0.853	94.0%	+0.140

Table 10: CRW performance and diagnostic benchmark metrics of each embedding model, sorted by composite rank in model selection (RQ2). $J_{36}^{\max}$ is the best mean Jaccard similarity reached over α for each clone type, averaged over the five clone types. Cell shading encodes each column’s rank independently, from best to worst: 1–2, 3–4, 5–6, 7–8, 9–10. The predictive metrics (topic coherence and negation invariance) track the CRW column closely, while ordering accuracy and correlation agreement show a more mixed pattern, reflecting their lack of predictive power (Table 11).

models. As visualized in Figure 13, topic coherence acts as an almost perfect predictor ($\rho_S = -0.915$, $p < 0.001$). Models that successfully form tight local clusters of same-topic variants relative to trap questions achieve dramatically better CRW performance. Negation invariance is the second-strongest predictor ($\rho_S = -0.733$, $p = 0.016$). Ordering accuracy, while positively associated, does not reach significance ($\rho_S = +0.624$, $p = 0.054$), indicating that correctly ordering global pairs matters less than compressing same-topic variants into a tight distance band.

Benchmark metric	ρ_S	p -value	Significant
Topic coherence ↓	−0.915	< 0.001	Yes
Negation invariance ↓	−0.733	0.016	Yes
Ordering accuracy ↑	+0.624	0.054	No
Correlation agreement ρ_S	+0.455	0.187	No

Table 11: Spearman rank correlation between diagnostic benchmark metrics and CRW performance $J_{36}^{\max}$ (Table 10) across the 10 embedding models. Topic coherence is the strongest predictor of CRW effectiveness ($\rho_S = -0.915$). Correlation agreement with d_{corr} does not predict CRW performance.

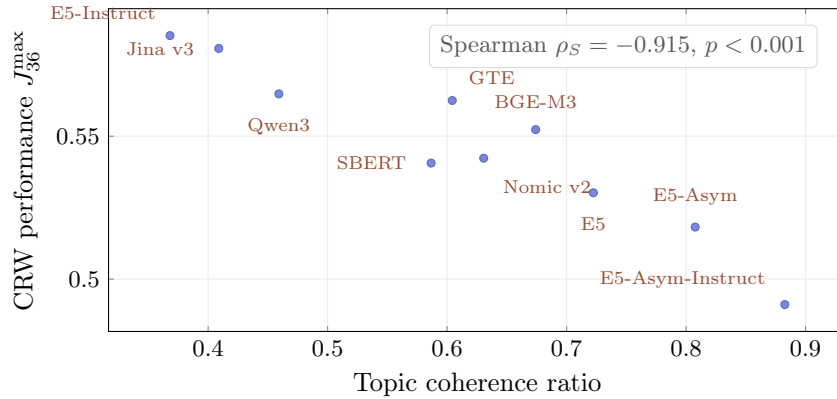


Fig. 13: Topic coherence ratio vs. CRW performance $J_{36}^{\max}$ (Table 10) across all 10 models. The strong negative correlation ($\rho_S = -0.915$, $p < 0.001$) indicates that models which successfully cluster functionally related questions tighter than deceptive keyword traps achieve dramatically better CRW performance. E5-Asym-Instruct is a clear case of this: despite high global ordering accuracy (94%), its poor local topic coherence makes it the worst-performing CRW model.

Correlation agreement does not predict CRW performance. Table 11 additionally reports each model’s Spearman agreement with the correlation-based distance matrix d_{corr} across all 2,775 question pairs. While all models exhibit a strictly positive rank correlation with empirical voter behavior, the absolute magnitude of this alignment is consistently weak, with individual model agreement scores ranging from $\rho_S = 0.060$ to $\rho_S = 0.198$. Figure 14 additionally visualizes this weak global alignment for E5-Instruct. Strikingly, a model’s baseline correlation agreement does not reliably predict its actual CRW effectiveness: when comparing these agreement scores against final CRW performance across the 10 models, the resulting meta-correlation is statistically insignificant ($\rho_S = +0.455$, $p = 0.187$). The top-performing model in baseline agreement, Qwen3, achieves a score of $\rho_S = 0.198$, while E5-Instruct (the definitively optimal model for CRW) ranks 7th with a mere $\rho_S = 0.105$.

E5-Asym-Instruct illustrates this dichotomy perfectly: it ranks 4th in global correlation agreement ($\rho_S = 0.140$) and achieves the 4th-highest global ordering accuracy (94.0%). Yet, it ranks last for CRW and possesses the worst local topic coherence (0.883) and negation invariance (0.853) of all 10 models, effectively failing to cluster related items tightly enough to trigger the defense.

D.3 Upper Bound with Behavioral Distances

To separate the limitations of CRW from those of the embedding, we repeat the full-questionnaire evaluation with the correlation-based distance $d_{\text{corr}}(q, q') = \arccos(|\rho(q, q')|)$ and exact copies as clones. Since clones share their original’s

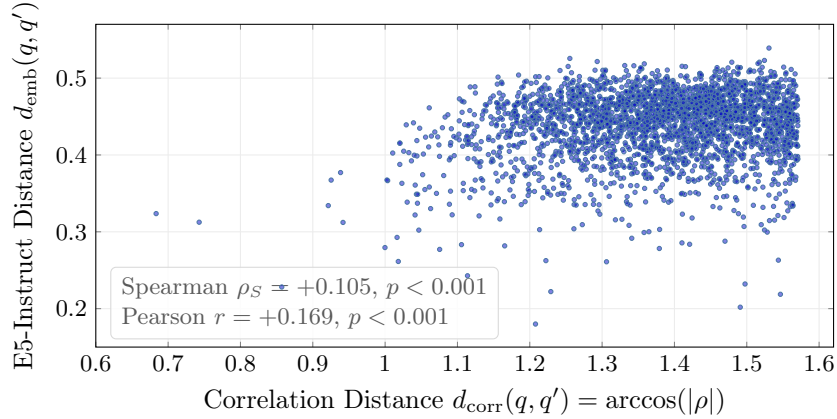


Fig. 14: Relationship between correlation distance $d_{\text{corr}} = \arccos(|\rho|)$ and E5-Instruct embedding distance across all 2,775 question pairs. The weak positive correlation (Spearman $\rho_S = +0.105$, $p < 0.001$) confirms that textual similarity and functional redundancy are largely decoupled: E5-Instruct captures topical and lexical proximity, while correlation distance reflects voter response patterns that are not reliably predicted by question wording alone.

response vector, they lie at distance zero, so a single clone condition suffices. As shown in Figure 15, CRW then achieves a near-perfect mean J_{36} of 0.992–0.998 over a wide plateau $\alpha \in [0.01, 1.5]$, with 71 of 75 questions above 0.99 at $\alpha = 1.5$, before degrading as distinct questions start sharing weight. Below the closest distinct pair ($d_{\min} = 0.68$, $|\rho| = 0.78$), J_{36} should equal 1 in exact arithmetic; the residual gap ($J_{36} = 0.992$, median 1) stems from equidistant candidates at the top-36 cutoff, whose ties are broken differently after CRW rescales the weights. These ties vanish once weights become non-uniform ($\alpha \geq 0.7$), where J_{36} rises to 0.998. The gap between E5-Instruct ($J_{36} = 0.870$ at $\alpha^* = 0.4$, averaged over clone types) and this ceiling indicates that the limiting factor is the embedding’s ability to group clones, not the CRW framework itself; the next subsection compares both distance structures in detail.

D.4 The Global Distance Landscape

We compare the pairwise distances between the 75 questions (2,775 pairs) under the E5-Instruct embedding and under the behavioral correlation distance $d_{\text{corr}} = \arccos(|\rho|)$ (Figure 16). Both distributions have an almost empty left tail: native questions are never near-duplicates. In embedding space, only 23 pairs lie below $d_{\text{emb}} = 0.3$, the closest at 0.18, while most pairs are packed between 0.3 and 0.5; at $\alpha^* = 0.4$, 463 pairs (16.7%) are within the CRW radius. In correlation space, only 8 pairs have $|\rho| > 0.54$, the strongest being $|\rho| = 0.78$, far from the $|\rho| \approx 1$ of actual clones; the mean absolute correlation is 0.19.

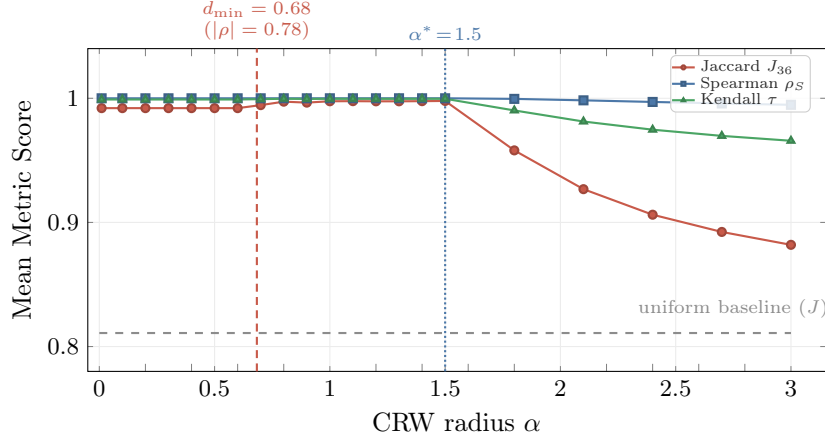
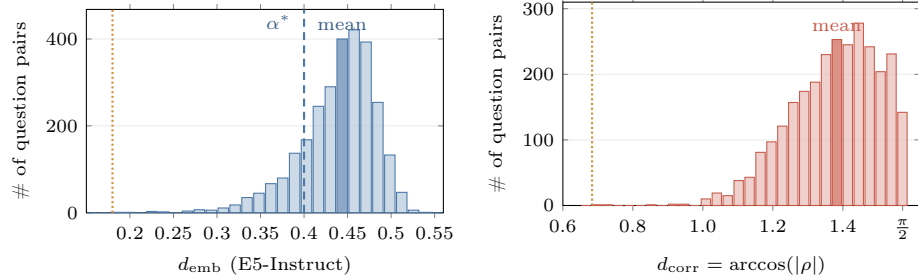


Fig. 15: CRW performance using the correlation-based distance metric $d_{\text{corr}}(q, q') = \arccos(|\rho(q, q')|)$ with identical clones, averaged over all 75 questions. CRW trivially achieves near-perfect correction ($J > 0.99$) across a wide α range of 0.01–1.5. The dashed line marks $d_{\min} = 0.68$, the distance of the closest non-clone question pair ($|\rho| = 0.78$). At $\alpha = 1.5$ performance starts drastically degrading. The residual imperfection below $d_{\min}$ (mean $J_{36} = 0.992$, median 1) stems from ties between equidistant candidates at the top-36 cutoff.



(a) E5-Instruct distance; dashed: $\alpha^* = 0.4$; (b) Correlation distance; dotted: closest pair (0.18). darker: closest pair (0.68, $|\rho| = 0.78$).

Fig. 16: Distribution of pairwise distances between the 75 questions (2,775 pairs). The darker bar contains the mean.

The two notions of proximity largely disagree (Table 12). Of the five closest pairs under each metric, only one is shared (Schengen and the Bilateral Agreements), and the closest embedding pairs are often weakly correlated: electricity-market liberalization and progressive tariffs, nearly adjacent in embedding space, rank 2,192nd of 2,775 in correlation. This mirrors the weak global agreement between the two distances ($\rho_S = 0.105$; Appendix D.2).

	Question pair	d_{emb}	$ \rho $	Rank
closest d_{emb}	Premium subsidies / health-care deductible (Q15/Q22)	0.180	0.355	300
	Electricity liberalization / progressive tariffs (Q42/Q51)	0.202	0.080	2,192
	US free trade / end Bilateral Agreements (Q70/Q72)	0.219	0.024	2,581
	Public-transport / road-traffic budget (Q84/Q85)	0.222	0.335	378
	End Schengen / end Bilateral Agreements (Q68/Q72)	0.228	0.655	3
highest $ \rho $	Army size / defence budget (Q64/Q87)	0.324	0.775	51
	Highway widening / road-traffic budget (Q45/Q85)	0.313	0.737	35
	End Schengen / end Bilateral Agreements (Q68/Q72)	0.228	0.655	5
	Public housing / social-services budget (Q19/Q81)	0.334	0.605	62
	Naturalization / foreigners' local vote (Q28/Q31)	0.367	0.602	187

Table 12: The five closest question pairs under the E5-Instruct distance (top) and under the correlation distance (bottom), with each pair’s rank (Rank) among the 2,775 pairs under the other distance. Question IDs abbreviate Q322xx (Appendix E); the shaded pair is the only one in both lists.

E Full Questionnaire

All 75 questions in the 2023 SmartVote questionnaire for the Swiss National Council elections, grouped by topic category. *Options* indicates the number of answer choices (4 = policy, 5 = budget, 7 = value). *Mini* indicates inclusion in the 30-question *rapide* questionnaire (✓).

Table 13: The 75 questions of the 2023 SmartVote questionnaire for the National Council elections (English version), grouped by SmartVote topic category. *Opt.* gives the number of answer options (4 for policy, 5 for budget, and 7 for value questions), and *Mini* marks the 30 questions of the short *rapide* questionnaire. Wording, including the [BePart question] tags, follows the original questionnaire.

ID	Question	Opt.	Mini
<i>Democracy & Media</i>			
Q32258	Should the Swiss mobile network be equipped throughout the country with the latest technology (currently 5G standard)?	4	
Q32259	Should the federal government be given additional powers in the area of digitization of government services in order to be able to impose binding directives and standards on the cantons?	4	
Q32260	Are you in favor of stronger regulation of the major Internet platforms (i.e., transparency rules on algorithms, increased liability for content, combating disinformation)?	4	✓
Q32261	A popular initiative aims to reduce television and radio fees (CHF 200 per household, exemption for businesses). Do you support this initiative?	4	✓
Q32262	Are you in favour of lowering the voting age to 16?	4	
Q32263	Should it be possible to hold a referendum on federal spending above a certain amount (optional financial referendum)?	4	
<i>Economy & Labour</i>			
Q32240	Are you in favor of introducing a minimum wage of CHF 4,000 for all full-time employees?	4	✓
Q32241	Do you support stricter regulations for the financial sector (e.g., stricter capital requirements for banks, ban on bonuses)?	4	✓
Q32242	Should private households be free to choose their electricity supplier (complete liberalization of the electricity market)?	4	
Q32243	Should housing construction regulations be relaxed (e.g., noise protection, occupancy rates)?	4	
Q32244	Are you in favor of stricter controls on equal pay for women and men?	4	✓
<i>Education</i>			

Continued on next page

Table 13 continued

ID	Question	Opt.	Mini
Q32225	According to the Swiss integrated schooling concept, children with learning difficulties or disabilities should be taught in regular classes. Do you approve of this concept?	4	
Q32226	Should the federal government raise the requirements for the gymnasiale matura?	4	
Q32227	Should the state be more committed to equal educational opportunities (e.g., through subsidized remedial courses for students from low-income families)?	4	✓
<i>Energy & Transport</i>			
Q32245	Should busy sections of highways be widened?	4	✓
Q32246	Should Switzerland ban the registration of new passenger cars with internal combustion engines starting in 2035?	4	✓
Q32247	To achieve climate targets, should incentives and target agreements be relied on exclusively, rather than bans and restrictions?	4	
Q32248	Do you think it's fair that environmental and landscape protection rules are being relaxed to allow for the development of renewable energy?	4	
Q32249	Should the construction of new nuclear power plants in Switzerland be allowed again?	4	✓
Q32250	Should the state guarantee a comprehensive public service offering also in rural regions?	4	
Q32251	Would you be in favor of the introduction of increasing electricity tariffs when consumption is higher (progressive electricity tariffs)?	4	
<i>Federal Budget</i>			
Q32281	Should the federal government spend more or less in the area of "Social services"?	5	
Q32282	Should the federal government spend more or less in the area of "Education and research"?	5	
Q32283	Should the federal government spend more or less in the area of "Agriculture"?	5	
Q32284	Should the federal government spend more or less in the area of "Public transport"?	5	
Q32285	Should the federal government spend more or less in the area of "Road traffic (motorised individual transport)"?	5	
Q32286	Should the federal government spend more or less in the area of "Public security"?	5	
Q32287	Should the federal government spend more or less in the area of "National defence"?	5	
Q32288	Should the federal government spend more or less in the area of "Development assistance"?	5	
<i>Finances & Taxes</i>			
Q32236	Do you support tax cuts at the federal level over the next four years?	4	✓

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Table 13 continued

ID	Question	Opt.	Mini
Q32237	Should married couples be taxed separately (individual taxation)? [BePart question]	4	
Q32238	Would you support the introduction of a national inheritance tax on all inheritances over one million Swiss francs? [BePart question]	4	✓
Q32239	Should the differences between cantons with high and low financial capacity be further reduced through fiscal equalization?	4	
<i>Foreign Policy</i>			
Q32269	Are you in favor of closer relations with the European Union (EU)?	4	
Q32270	Should Switzerland strive for a comprehensive free trade agreement (including agriculture) with the USA?	4	✓
Q32271	Should Swiss companies be obliged to ensure that their subsidiaries and suppliers operating abroad comply with social and environmental standards?	4	
Q32272	Should Switzerland terminate the Bilateral Agreements with the EU and seek a free trade agreement without the free movement of persons?	4	✓
Q32273	Should Switzerland return to a strict interpretation of neutrality (renounce economic sanctions to a large extent)? [BePart question]	4	✓
<i>Health</i>			
Q32220	Should compulsory vaccination of children be introduced based on the Swiss vaccination plan?	4	
Q32221	Are you in favor of the introduction of a tax on foods containing sugar (sugar tax)?	4	
Q32222	Should insured persons contribute more to health care costs (e.g., increase the minimum deductible)?	4	✓
Q32223	Should the Federal Council's ability to restrict private and economic life in the event of a pandemic be more limited?	4	✓
Q32224	Should the federal government be given the authority to determine the hospital offering (national hospital planning with regard to locations and range of services)?	4	
<i>Immigration</i>			
Q32228	Should the conditions for naturalization be relaxed (e.g., shorter residence period)?	4	✓
Q32229	Should more qualified workers from non-EU/EFTA countries be allowed to work in Switzerland (increase third-country quota)?	4	
Q32230	Do you support efforts to house asylum seekers in centers outside Europe during the asylum procedure?	4	✓
Q32231	Should foreign nationals who have lived in Switzerland for at least ten years be granted the right to vote and stand for election at the municipal level?	4	✓

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Table 13 continued

ID	Question	Opt.	Mini
<i>Nature Conservation</i>			
Q32252	Are you in favor of further relaxing the protection regulations for large predators (lynx, wolf, bear)?	4	✓
Q32253	Should direct payments only be granted to farmers with proof of ecological performance?	4	✓
Q32254	Are you in favour of stricter animal welfare regulations for livestock (e.g. permanent access to outdoor areas)?	4	
Q32255	Should 30% of Switzerland's land area be dedicated to preserving biodiversity? [BePart question]	4	
Q32256	Would you support a ban on single-use plastic and non-recyclable plastics? [BePart question]	4	✓
Q32257	Are you in favour of government measures to make the use of electronic devices more sustainable (e.g., right to repair, extension of warranty period, minimum guaranteed period for software updates)? [BePart question]	4	
<i>Security & Military</i>			
Q32264	Are you in favor of expanding the army's target number of soldiers to at least 120,000?	4	✓
Q32265	Should the Swiss Armed Forces expand their cooperation with NATO?	4	
Q32266	Should the Federal Council be allowed to authorize other states to re-export Swiss weapons in cases of a war of aggression in violation of international law (e.g., the attack on Ukraine)? [BePart question]	4	
Q32267	Should automatic facial recognition be banned in public spaces?	4	✓
Q32268	Should Switzerland terminate the Schengen agreement with the EU and reintroduce more security checks directly on the border? [BePart question]	4	✓
<i>Society & Ethics</i>			
Q32232	Should cannabis use be legalized? [BePart question]	4	✓
Q32233	Would you be in favour of doctors being allowed to administer direct active euthanasia in Switzerland?	4	
Q32234	Should a third official gender be introduced alongside "female" and "male"?	4	✓
Q32235	Do you think it's right for same-sex couples to have the same rights as heterosexual couples in all areas?	4	
<i>Values</i>			
Q32274	What is your position the following statement: "Someone who is not guilty, has nothing to fear from state security measures."	7	
Q32275	What is your position the following statement: "In the long term, everyone benefits from a free market economy in the long term."	7	

Continued on next page

Table 13 continued

ID	Question	Opt.	Mini
Q32276	What is your position the following statement: “It is necessary for the State to balance out differences in income and wealth through redistribution”.	7	
Q32277	What is your position the following statement: “It is best for a child, when one parent stays home full-time for childcare.”	7	
Q32278	What is your position the following statement: “The ongoing digitalization offers significantly more opportunities than risks.”	7	
Q32279	What is your position the following statement: “Punishing criminals is more important than reintegrating them into society.”	7	
Q32280	What is your position the following statement: “Stronger environmental protection is necessary, even if its application limits economic growth.”	7	
<i>Welfare & Family</i>			
Q32214	Do you support an increase in the retirement age (e.g., to 67)?	4	✓
Q32215	Should the federal government allocate more funding for health insurance premium subsidies?	4	✓
Q32216	For married couples, the pension is currently limited to 150% of the maximum individual AHV pension (capping). Should this limit be eliminated?	4	
Q32217	As part of the reform of the BVG (occupational pension plan), pensions are to be reduced (lowering the minimum conversion rate from 6.8% to 6%). Are you in favor of this measure?	4	
Q32218	Should paid parental leave be increased beyond today’s 14 weeks of maternity leave and two weeks of paternity leave? [BePart question]	4	
Q32219	Should the federal government provide more financial support for public housing construction? [BePart question]	4	✓